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Evaluation Robustness of Climate NLP Benchmarks: Revising SciDCC through Lexical Transformations in Multi-Label Scenarios

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Abstract

Large language models are increasingly applied to climate-related text analysis, yet existing climate benchmarks have been shown to contain annotation errors, inconsistencies, and design weaknesses. These issues cause them to fail to adequately capture model performance and raise questions about the reliability of current evaluation practices in the climate change domain. This study analyses the Science Daily Climate Change (SciDCC) benchmark for single-label topic classification and proposes a revised multi-label classification and ranking framework that introduces robustness testing through lexical transformations on a word and sentence level. We generate three different text variants, involving paraphrasing, masking, and code-mixing strategies, and evaluate four large language models – Mistral-Large 2 (Mistral AI), Llama3.3 70B (Meta), GPT-OSS 20B (OpenAI), and Llama3.2 3B (Meta) on the extended benchmark through label and ranking overlap, consistency, robustness, inter-model analysis, and human annotation. The results show that the redesigned multi-label classification task is well-defined, and models remain highly robust under text variations, although the transformations are partially limited in exposing the differences among the top-performing models. The ranking task exhibits lower inter- and intra-annotator agreement and greater disagreement between models and annotators. Nonetheless, robustness scores remain high across most transformations, which may be attributed to certain limitations of the evaluation metrics. The study contributes a first attempt at robustness and consistency evaluation in the climate domain and suggests strategies for further benchmark improvement and task refinement.

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Preface

This thesis marks the end of my Master's studies, a period that had it all – the excitement, teamwork, friendships, and new impressions, along with the inevitable occasional frustration and tiredness that ultimately turn back into curiosity and determination. A happy, memorable time.

If we follow the chain of events, the beginnings of this time might well have been over ten years ago, though that seems to be an arbitrary line to draw, too. Regardless, I would like to extend my sincerest gratitude to everyone involved, and special thanks to several people who played the most crucial parts.

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To whoever is reading this, thank you for your time, and I would greatly appreciate your generous feedback.

1 Introduction

Over the past few decades, climate change has transitioned into a major topic in the scientific, political, and social discourse, as visible environmental impacts and growing concern for ecological issues have brought it out of obscurity. At the same time, natural language processing (NLP) methods are increasingly being applied to process and analyse large amounts of textual data across various domains. As more and more environmental reports, media content, and scientific literature on climate change become available, NLP systems are being used to classify, interpret, and assess the climate-related texts and apply insights from them in research and the industry. However, the emerging use of models and applications also creates a need for evaluation frameworks that can reliably measure how well the systems perform on climate-domain tasks. In this thesis, we examine how NLP systems are evaluated on climate-related data and identify limitations in an existing benchmark.

Recent trends in climate-domain benchmarking have prioritised speed and scalability, often resulting in datasets that are only partially curated by human experts (Manivannan et al., 2025). While this tendency for benchmark mass production is understandable and reflects similar developments across other data-intensive fields, it raises concerns for the quality of model evaluation in climate-related research. Several existing climate benchmarks have been shown to exhibit serious weaknesses (Calamai et al., 2025): a majority of them contain annotation errors and inconsistent labels in their ground-truth data, while others have been crowd-annotated without any clear guidelines; some benchmarks are limited by static designs that fail to adequately capture the versatility and complexity of the test data. Moreover, benchmark results may contain biases and are often sensitive to small changes in wording, even when the semantic meaning of a question remains the same. Models evaluated on these benchmarks demonstrate unstable performance, particularly in classification tasks involving many categories. These issues undermine confidence in existing benchmarks as trustworthy measures of models’ performance and reasoning, and call attention to the reliability, robustness, and interpretability of current evaluation practices. There is, therefore, a growing demand for evaluation frameworks that maintain efficiency without compromising on quality.

Benchmarks in the climate domain have the added difficulty of a constantly shifting field with emerging topics and controversial political and social rhetoric that continuously reshape the data distribution and contribute to benchmark instability. At the same time, rapid advances in NLP and the evolving abilities of language models further challenge the evaluation process, as benchmark designs and annotation frameworks can quickly become outdated or insufficiently challenging to measure model performance.

To address these limitations, we analyse in depth the Science Daily Climate Change (SciDCC) benchmark (P. Mishra and Mittal, 2021) and study annotation artifacts and labelling inaccuracies by comparing outputs from several baselines and state-of-the-art large language models (LLMs). We aim to tackle the identified issues by transforming the benchmark’s task from single-label topic classification by article body and title to multi-label topic classification and ranking. Furthermore, we include a robustness evaluation component to assess models’ relative behaviour under lexical transformations on a word and sentence level, including paraphrasing, code-mixing,

and masking strategies. We examine the performance and consistency of multiple LLMs across the transformed inputs for both redesigned tasks and compare model predictions with human annotations.

We seek insight into the following research questions:

- RQ1: How well does a multi-label classification and ranking benchmark task design capture the underlying data complexity and labelling noise?
- RQ2: How consistent are model predictions under input transformations that preserve semantic meaning? How do relative performance trends between models change across those variations?
- RQ3: How consistent are models' multi-label classification and ranking outputs with human annotations?

To our knowledge, this is the first attempt to develop a robustness evaluation framework for climate-related benchmarks. By incorporating transformed text variants in the evaluation process, we aim to move beyond single-label correctness as the main evaluation metric and explore ways to utilise weakly labelled datasets without extensive manual re-annotation. The study contributes an extended benchmark for multi-label topic classification and ranking downstream tasks and suggests strategies for further benchmark improvement and task refinement adapted to the challenges of climate NLP.

The thesis has the following structure: Chapter 2 introduces key concepts and definitions in (climate) NLP benchmarking, including data annotation, weak and strong labels, static and dynamic benchmarks, and robustness evaluation, and provides an overview of current research directions and findings. Next, Chapter 3 lays out the methods and data used throughout the study, along with the models and baselines we employ to evaluate the benchmark tasks, as well as the text transformations and the annotation procedure. The results of the study are presented in Chapter 4 and further interpreted in Chapter 5. Finally, Chapter 6 summarises the findings and limitations of the thesis, answers the research questions, and suggests directions for future research and improvement.

2 Background

A benchmark is a standardised dataset with corresponding evaluation methods designed to assess and compare the performance of language models on a variety of natural language processing (NLP) tasks, such as text classification, question answering, claim detection, entity recognition, machine translation, etc. There are many different aspects of benchmark creation and implementation, including task design and complexity, dataset curation, annotation quality, evaluation metrics, real-world applicability and efficiency, bias, fairness, and reproducibility, all to ensure that benchmarks reflect models’ progress, strengths, and limitations and evolve along with them.

This chapter presents a brief overview of general NLP benchmarks and climate-domain benchmarks and their properties and challenges. We define several key terms that will be used throughout this work: weak and strong labels, static and dynamic benchmarks, robustness, and test transformations/perturbations.

2.1 NLP Benchmarks

NLP benchmarks have evolved from a long tradition of evaluation methods, beginning in the 1950s and 1960s with the establishment of core metrics like precision and recall, followed by the F1-score in the 1970s, to measure the quality of retrieved information (Sokolova et al., 2006, Banerjee et al., 2024). Early attempts to evaluate language models in specialised tasks included key metrics such as BLUE (Papineni et al., 2002) and ROUGE (Lin, 2004), both measuring surface-level n-gram overlap in machine translation, as well as perplexity (Jelinek, 1977), which was originally introduced in the context of speech recognition and indicates how well a probability model predicts a sample.

Building on task-specific metrics, NLP evaluation shifted towards single-task benchmarks, designed to measure an individual linguistic property through one or several key metrics. Some widely used examples include the shared task of CoNLL-2003 (Tjong Kim Sang and De Meulder, 2003) for language-independent named entity recognition, the IMDB dataset (Maas et al., 2011) of positive and negative movie reviews for binary sentiment classification, and the Penn Treebank Project (Marcus et al., 1993) – a corpus, used to train and evaluate syntactic parsers and part-of-speech taggers. While these benchmark datasets provide an excellent, standardised base for comparison across both statistical and deep-learning language models, they often remain bound to a single task and to a limited set of metrics.

Since then, benchmark evaluations have transformed into sizable frameworks: The General Language Understanding Evaluation (GLUE) benchmark (A. Wang et al., 2018) was one of the first multi-task benchmarks that pushed towards a more comprehensive, model-agnostic evaluation by combining multiple task-specific metrics across diverse NLP tasks and making a versatile range of text genres, dataset sizes, and degrees of difficulty available for testing. Building upon this approach, SuperGLUE (A. Wang et al., 2019) introduces more complex tasks that require deeper language understanding, while MMLU (Hendrycks et al., 2021) measures multitask accuracy, covering 57 tasks and testing extensive world knowledge and models’ problem-solving.

Benchmarks and evaluation frameworks are meant to serve as the primary reference point for language models’ abilities in language understanding, reasoning, and

generation. However, due to factors like data contamination and continuous LLM improvement, benchmarks are becoming saturated and outdated, facing more and more limitations. Models are increasingly achieving near-perfect scores on many well-established benchmarks, but fail at genuine language understanding and adaptability, raising concerns about data leakage, research integrity, annotation bias, and benchmarks’ scope (Banerjee et al., 2024). The following subsections present some key properties, challenges, and mitigation strategies related to NLP benchmarks and this thesis project.

2.1.1 Data Annotation

Annotation quality is crucial for the evaluation validity of benchmarks. Benchmark datasets are designed to compare models’ performance on one or several tasks and rely on a solid base for comparison. Regardless of how annotations have been collected – whether through human annotators, automated procedures, or hybrid approaches – they can introduce noise, bias, and inconsistencies that might alter the evaluation outcomes.

Human-as-a-Judge methodologies have proven effective, in particular in benchmarks like DynaBench (Kiela et al., 2021), which makes use of human annotators to produce adversarial examples and to determine the level of understanding and creativity of the output. Models are then tested and evaluated based on their responses to the adversarial examples. The approach provides a more robust evaluation, as it captures real-world complexity and model weaknesses that automated metrics might fail to detect.

Despite the conceptual soundness of such frameworks, NLP benchmarks are still being ‘hacked’. Two of the reasons for that are annotation artifacts (Gururangan et al., 2018) and annotator bias (Chiang et al., 2024; Gevers et al., 2025). Gururangan et al. (2018) show that annotation artifacts are common and highly correlated to the framing of the annotation task: The authors experimented with having crowd workers generate hypotheses and found that the data created in the process contained linguistic patterns, which can help models classify premise–hypotheses pairs correctly without ever having seen the premises. Furthermore, human annotators often exhibit inconsistent preferences towards certain output formatting, formality, or tone, and differ in their annotation styles (Chiang et al., 2024). The bias can be detected and exploited by models, resulting in unreliable scores instead of genuine language understanding. One way to mitigate these effects is to augment the data by, for instance, automatically generating perturbed versions of the test instances, in such a way that the perturbations are not related to the original task (Gevers et al., 2025). Benchmarks could also be filtered to exclude contaminated, too easy or biased examples (Chiang et al., 2024; Gevers et al., 2025).

Furthermore, reasons for diminished annotation quality could be annotator disagreement, which might be grounded in the clarity of the guidelines, the skill level of the annotators, or the complexity of the data (Gururangan et al., 2018). Mitigation strategies include improving the guidelines, preserving individual annotator responses rather than collapsing them into a single aggregated label, or including more granular forms of annotation (Gevers et al., 2025), such as asking annotators to respond on a scale rather than provide a single, unnuanced label. These solutions reflect a perspective on annotation that emphasises the importance of diverse annotator perspectives in NLP tasks and treats divergence and disagreement as informative signals rather than noise (Gruber et al., 2024).

Another approach to data annotation utilises large language models (LLMs) to generate or evaluate outputs. The LLM-as-a-Judge methodologies bring a new set

of challenges, such as bias related to the model’s fine-tuning and training data, self-preference bias, where models rate their own outputs higher than human annotations and other models’ outputs (Banerjee et al., 2024), sensitivity to prompt formulations, and inability to provide confidence scores and judge their own responses. Hybrid methods aim to combine the strengths of human and automated annotation and to find a good balance between low cost and high quality. However, these approaches fall outside of the scope of this work and are therefore not discussed further.

2.1.2 Weakly and Strongly Labelled Datasets

The distinction between weak and strong labels originates from the weak supervision literature, where strong labels refer to human annotations (most often, by experts) and are frequently treated as gold-truth labels. They are expensive to obtain – both in annotator-hours and financial cost – but, presumably, of high quality. Weak labels, on the other hand, are generated through user-written heuristics, such as assigning labels to data points if they contain a keyword or match the specified regular expression, or by leveraging another model to assign labels based on pre-defined criteria (J. Zhang et al., 2022). In some cases, annotations by non-expert humans might also be considered weak labelling (Snow et al., 2008). Weak labels, as the name suggests, lack the strong definitions of manually assigned labels and are noisy and less reliable for classification tasks. They can introduce inconsistencies if, for instance, label definitions evolve over time. On the benefits side, they are cheap and fast to produce, easily scalable for large datasets without requiring human annotations, and can be used as an approximate estimation of the ground truth.

In this project, we work with both weakly and strongly labelled data. We attempt to improve a weakly labelled benchmark dataset by re-designing its core task to fit more closely with the nature of the data and to mitigate the downsides of the noisy and, as we show, often incorrectly assigned labels. Subsequently, we use a small subset with strong labels to evaluate our results.

2.1.3 Static vs. Dynamic Benchmarks

Benchmarks can be categorised into static and dynamic, depending on whether benchmark datasets remain fixed or evolve constantly. Static benchmarks consist of definite, standardised datasets that allow for easy comparison, consistency, and reproducibility of results across different models and their versions over time. They are stable, with well-defined tasks and efficient error analysis. Often, traditional static benchmarks like GLUE and MMLU are considered the gold standard when models are released and tested to determine their position on leaderboards.¹

However, static benchmarks also have limitations that require solutions, as new, bigger, and more capable models are being deployed. For one, the datasets are to a large extent publicly available, which allows for the data to be used in model training – a problem known as ‘data contamination’ or ‘test data leakage’. If models are exposed to the benchmarks in their training data, they may memorise data-specific patterns rather than acquire true language understanding, leading to unrealistically high results and overconfidence when, in reality, the same models might fail to accomplish the task. Moreover, as both open-source and commercial models compete to achieve state-of-the-art performance, they are increasingly optimised for specific benchmarks without actually improving (Boubdir et al., 2023).

¹For example, the Open LLM Leaderboard by Hugging Face: <https://huggingface.co/open-llm-leaderboard>.

Dynamic benchmarks attempt to circumvent the limitations of static benchmarks by continuously updating the dataset and generating new test cases during evaluation based on the performance of the model. They are set to anticipate dataset saturation and to ‘recycle’ easy datasets, as opposed to the production of new benchmarks that would soon become outdated. A study on dynamic benchmarking for code language models (Guan et al., 2025) finds that automatically transformed test inputs, which are syntactically new but semantically identical, cause an up to 40% performance drop, successfully working against the data contamination problem. Other frameworks like DynaBench use human annotators and models to dynamically create challenging examples that a target model will mislabel when a person would not. The difficulty level of the annotated examples is dynamically adapted to better models.

The dynamic approach, while successfully addressing the pitfalls of static benchmarking, is expensive and time-consuming, as dynamic benchmarks require continuous maintenance, updating, and annotations. Since the test set is renewed frequently, comparisons over time can be less reliable, and results are difficult to reproduce. Last but not least, the approach hides the risk of distributional shifts and lack of representation due to how the data is generated (Shirali et al., 2023). Nevertheless, current benchmarking attempts tend to be oriented towards more adaptive and actively developing frameworks to reflect increasing model complexity and evaluation demands.

2.2 Robustness Benchmarking

In the context of language models, robustness generally refers to ‘the model’s ability to maintain consistent performance, reliability, and adherence to intended behaviour (including accuracy, factuality, safety, and fairness) despite variations in inputs, contexts, or underlying data distributions. This encompasses resilience to natural noise, semantic paraphrasing, distribution shifts, incomplete information, reasoning perturbations, instruction variations, and adversarial manipulations’ (Kumar and S. Mishra, 2025). Accordingly, adversarial manipulations (also called ‘attacks’) are considered as inputs designed to mislead or confuse the model into producing incorrect or intended outputs, despite the outputs appearing normal to humans (Kumar and S. Mishra, 2025). In this work, robustness relates to the consistency and stability of outputs under systematically crafted text transformations.

Several studies of robustness and reliability of benchmark evaluations have found that even though benchmarks offer reasonable rankings of models (provided that datasets have not leaked into the training data), they also tend to overestimate absolute model performance and generalisation capability (Banerjee et al., 2024; Lunardi et al., 2025). Since the research of language models is not only concerned with how models compare to one another, but also with their actual ability to understand and produce human-readable language, robustness benchmarking has become essential for assessing models’ vulnerability to challenging, deceptive, or malicious inputs.

Robustness benchmarks can be broadly divided into datasets with automatically generated perturbations and human-in-the-loop frameworks. Automated approaches offer scalability, and human-driven methods usually produce more diverse and realistic adversarial examples. As with standard benchmarks, robustness datasets can target one or multiple tasks. We focus on two sequential tasks: multi-label topic classification and ranking, and apply an automatic approach to produce the perturbed variants.

Text Transformations for Robustness Evaluation A common way to evaluate robustness in NLP benchmarks is through systematic text transformations, which

introduce controlled noise into the dataset. Their purpose is to expose weaknesses and data leakage in language models, as well as to create challenging inputs and test the models’ language understanding by identifying whether or not the models rely on surface-level cues and pre-learned patterns to complete a task successfully. One further goal is to simulate real-world scenarios, where part of the data is corrupted – for instance, when it contains typos.

There are different taxonomies of transformations, and a common categorisation approach is by the NLP task that the inputs will be used for. Frameworks like Textflint (X. Wang et al., 2021), for instance, include both general and task-specific transformations. Many studies on adversarial robustness through perturbations distinguish between transformations on a character, word, and sentence level. Character-level transformations include different types of shuffling, disemvoweling, truncating, segmentation, keyboard and natural typos, etc. (P. Wang et al., 2025). To word transformations belong composition-, knowledge-, context-, and embedding-based perturbations, while some benchmarks like AdversarialGLUE (B. Wang et al., 2021) also add typos to the word-level class. Sentence-level transformations are executed through manipulation of multiple words (seen as an advanced form of word-level perturbations) or by restructuring sentences based on specific grammatical rules; they are more flexible than the other types, as changes can be made anywhere in the sentence. AdversarialGLUE defines a fourth category of perturbations: human-generated examples, which are to capture more versatile linguistic phenomena in addition to the automatic algorithms.

Like other benchmark types, robustness benchmarks face limitations. They are often criticised for including perturbed inputs that do not preserve the meaning or are unnatural to humans and are therefore not applicable in real-world settings where people are involved. Dyrmishi et al., 2023 analyse human perception of word replacement attacks and conclude that a substantial amount of adversarial examples would not pass the human judgment barrier, raising questions about the method’s usefulness under real deployment. However, the validation scheme proposed in the study requires a high number of human annotators to screen and assess examples, which, in turn, is impractical in large-scale projects. These limitations emphasise a trade-off in robustness evaluation through textual and linguistic transformations: Human validation can improve the level of realism of adversarial examples, but it can also introduce subjectivity, variability, and scalability constraints.

Ultimately, the choice of perturbations and the perturbations’ generation method depends on the task requirements and the evaluation goals. The transformations applied in this work are aligned with the aim of the study, the dataset characteristics, and the amount of available resources; accordingly, we perform automatic lexical transformations on a word/sentence level and assess their quality by analysing a subset of each generated variant.

2.3 Climate NLP Benchmarks

Climate NLP benchmarks mirror the design principles and evaluation pitfalls of general NLP benchmarks; however, they are tightly bound to the needs and constraints of the climate domain.

Climate benchmarking is an emerging field that owes its rapid development to the increasing amounts of climate-related data and the growing demands of research, society, and politics for automated methods in the climate change domain. In recent years, the more and more evident impacts of environmental change and crisis have led to the production of new climate data resources, such as scientific and environmental agencies’ reports, books, news, social media posts, and policy

documents. Subsequently, climate-specific NLP tasks like green claims identification and climate-related discourse detection have emerged to process, classify, and evaluate that data. As general benchmarks fail to capture climate-specific language and tasks effectively, developers of dedicated climate NLP benchmarks have stepped in to enable more meaningful evaluation.

ClimaBench (Spokoyny et al., 2023) was one of the first attempts to consolidate multiple datasets into a unified NLP benchmark for climate-related tasks. It includes several existing datasets and covers a range of tasks and data types: Climate-Stance for stance classification, ClimaText for sentence classification, Climate-Eng and SciDCC for topic classification, and Climate-FEVER for claim verification. Additionally, it introduces three further datasets: ClimaINS and ClimaQA for question answering and ClimaTopic for topic classification. Five of the datasets contained in ClimaBench were among the benchmarks used to evaluate the ClimateGPT model family (Thulke et al., 2024) – specialised LLMs that were fine-tuned for climate change and sustainability topics.

Alongside ClimaBench, several other benchmarks have been proposed to evaluate models on aspects of the climate change discourse. ClimateEval (Kurfali et al., 2025) is a collection of 13 datasets covering 25 different tasks, including some more specialised datasets like Climate-Change-NER for entity recognition, PIRA for reading comprehension (also included in the ClimateGPT evaluation), CheapTalk for climate detection, sentiment, commitments, and specificity, and others. Moreover, ClimateEval introduces the Guardian Climate News Corpus – a dataset for topic classification from web-scraped news articles.

The variety of task types covered by these benchmarks reflects the different ways in which climate-related tests can be analysed. However, climate-domain benchmarks face a lot of challenges too. For one, the climate discourse is very dynamic, and political and social burning topics are constantly shifting, causing context ambiguity, evolving vocabulary and connotations, and, on the practical side, resource allocation uncertainty. For another, climate benchmarks are confronted with the same limitations as general benchmarks in terms of annotation reliability, bias, and saturation. To successfully work around these issues, evaluation setups need to account for the ever-shifting language, theme, and landscape complexity of the climate domain.

Evaluation in climate NLP benchmarks generally uses macro-averaged F1 scores to measure model performance, as it enables consistent comparison across models, datasets, and tasks. Since it does not explicitly capture how models behave under input variation, we also include robustness-oriented evaluation alongside the standard metrics.

Calamai et al. (2025) summarise the findings of a study on 29 climate-related benchmarks’ quality issues, testing 6 models in 8 different tasks. The authors evaluate several baselines, including traditional machine learning systems like TF-IDF with logistic regression and fine-tuned Transformer models, as well as open-source and commercial LLMs; thereafter, they perform a detailed analysis of manually annotated errors. The results show that the TF-IDF, fine-tuned models, and LLMs in a zero-shot setting reach a similar performance. For classification tasks, this implies that classes are distinguishable using mostly surface-level cues that a simple logistic regression model could capture just as well. The finding raises the question of whether benchmarks are complex enough to test the language understanding of advanced models, and if LLMs also focus on vocabulary patterns. Surprisingly, fine-tuned models consistently outperform the zero-shot approaches, and for weakly labelled datasets such as the SciDCC and ClimaTopic, the performance gap is statistically significant. That could be explained by the fact that both datasets were produced automatically and have not been annotated by humans; constructing a comprehensive prompt for an LLM is

less intuitive and yields inconsistent results, while the fine-tuned Transformer models operate based on patterns in the training data and therefore remain uninfluenced in identifying the relative linguistic patterns.

A further discovery of the study is related to the internal quality of the datasets: 96% of the reviewed benchmarks contain annotation issues, with over 16% of the sampled incorrect predictions being direct annotation mistakes and over 38% being ambiguous examples that could belong to multiple categories or be discarded as not belonging to any available class (Calamai et al., 2025). The main problems outlined here are inconsistent annotations or the lack thereof, and inadequate benchmark design.

This thesis builds on the limitations presented above and attempts to address some of them by proposing a method to robustify weakly labelled climate benchmarks. We approach the issues at both the task design level and the single data point level, and highlight the possibility of dynamically updating the dataset and adding new challenging alterations to the original data. The methods, data, and models used are presented in Chapter 3.

3 Methodology

The following chapter presents the methodology applied throughout this study. Generally, our experimental approach is divided into three main components: an exploration phase, where we test and analyse the benchmark to uncover potential weaknesses and errors; an experimental phase, in which we transform the original dataset and task and evaluate several models’ relative performance under text perturbations; and a validation phase that includes the process of annotating a subset of the data and comparing the annotated examples to the model predictions.

Benchmark Exploration and Single-Label Classification In the first phase of our study, we reproduce the results by Calamai et al., 2025, testing three baseline models (a random baseline, a fine-tuned Transformer, and a TF-IDF logistic regression model) and three large language models (Mistral-Large-2-Instruct-2411, Llama3.3-70B-Instruct, and GPT-OSS-20B) in a zero-shot setting on one of the original single-label classification tasks – topic classification by article body. Subsequently, we perform a detailed manual error analysis on 100 randomly sampled examples, processed by seven different LLMs (Mistral-Medium-3.1, Mixtral-8x22B, Mistral-Large-2-Instruct-2411, Llama3.3-70B-Versatile, GPT-OSS-20B, GPT-OSS-120B, and Qwen3.5-397B).

Transformations Generation and Experiments with Multi-Label Classification and Ranking Building on the findings from the error analysis, in the second part of the study, we extend the benchmark’s single-label topic classification task by introducing two additional tasks: multi-label classification and ranking. We use the original article bodies as input and first prompt the models in a zero-shot setting to select a set of labels that fit the article’s topic, and then to rank them by their relevance to the text.

Instead of attempting to re-annotate the entire dataset to assess the new tasks, we aim to reduce the benchmark’s dependence on single label(set)s being truthful by transforming it into a diagnostic framework that measures model consistency and robustness under text transformations. To achieve this, we generate three variants of the original dataset (paraphrased, with masked keywords, and code-mixed) and evaluate each LLM on all datasets in a zero-shot setting. The experiments are conducted in two separate steps to ensure high-quality outputs and a clear distinction between the multi-label classification and the ranking tasks. We track relative performance and stability across models and variants.

Annotation Procedure and Results Evaluation The third part of the study addresses the question of whether relative model consistency under transformations is indicative of prediction correctness. We have a randomly sampled subset of the dataset annotated by human annotators and compare the models’ predictions to the annotations.

3.1 Data

As our target benchmark, we use the Science Daily Climate Change dataset (SciDCC) (P. Mishra and Mittal, 2021), which includes 11 539 unique climate change news articles from the period 1997–2021, web scraped from the ‘Earth & Climate’ and ‘Plants & Animals’ topics in the environmental science section of the Science Daily website.¹ Each article in the dataset consists of a title, a summary, and a text body. The SciDCC dataset was initially released to serve as source material for extracting named entities for knowledge graph construction in the climate change domain (P. Mishra and Mittal, 2021). Since each article was published under a specific news category and that corresponding category was extracted along with the article and assigned as its label, the dataset could be repurposed for several text classification tasks. The original classification tasks include the articles’ titles, bodies, summaries, and combinations of the three as input, and expect the respective category as output. In our experiments, we work with the articles’ bodies and associated categories. The average body length is 532.2 words, and the maximal body is 1968 words long.

The SciDCC dataset includes 20 categories, covering a variety of environment-related topics, such as ‘Weather’, ‘Pollution’, ‘Global Warming’, etc. There is a maximum of 1k news articles for each category. A full list of the categories and their distribution in the original dataset is presented in Figure 3.1. The dataset is highly imbalanced, and some categories are under-represented (for instance, ‘Global Warming’ appears only 21 times overall). We treat the categories as classification labels. The dataset is weakly labelled because the extracted article–category tag pairs have not undergone any manual correction or annotation procedures after the web scraping process. The articles were originally placed under the thematic categories by the editorial team at Science Daily, but to our knowledge, no further information on the category assignment criteria has been made available.

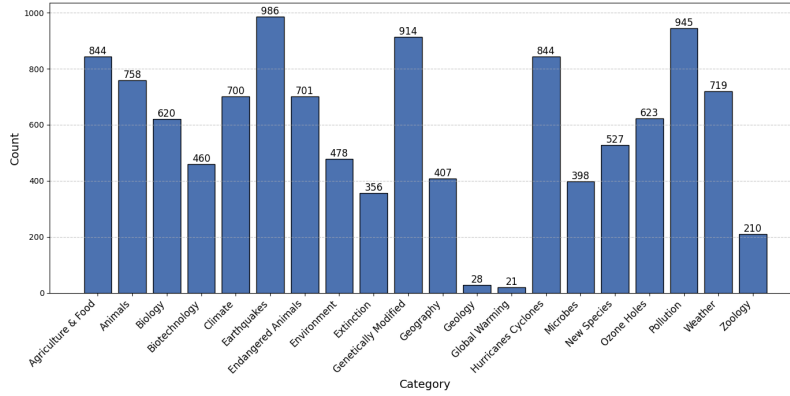


Figure 3.1: Label distribution in the SciDCC dataset (alphabetical order).

There are no official splits into training and test data provided. We split the dataset into 80% train, 10% validation, and 10% test data (train size: 9231 articles, validation size: 1154 articles, test size: 1154 articles) and use the same training and validation splits for the TF-IDF classifier and the fine-tuned DistilRoBERTa model and the same test split across all methods. Figure 3.2 presents the distribution of the training, validation, and test data. We use an identical fixed random seed (16) across all data splitting and sampling processes to ensure reproducibility and consistent data selection throughout all experiments.

¹<https://www.sciencedaily.com>.

3.2 Baselines and Models

The following section describes the models and the baselines used in our experiments. We construct three supervised baseline classifiers to reproduce Spokoyny et al. (2023)’s and Calamai et al. (2025)’s results on the single-label classification task as a foundation for the error analysis and the benchmark exploration. We evaluate three LLMs in a zero-shot setting on the original task and further include a fourth LLM as a baseline on the redesigned multi-label tasks. Finally, we consider a range of large language models with differing architectures, numbers of parameters, and inference setups to identify potentially problematic examples in the dataset and use additional models to generate the text variants.

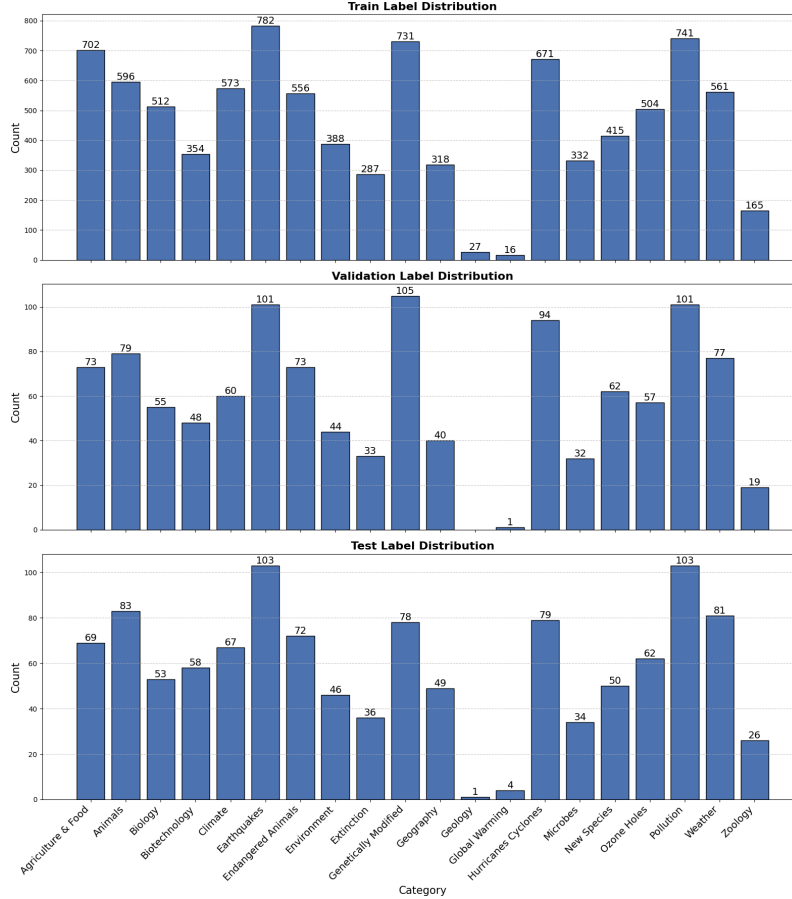


Figure 3.2: Label distribution of the train, validation, and test splits created from the SciDCC dataset (alphabetical order).

3.2.1 Random Baseline

We compute a random baseline using Sklearn’s `DummyClassifier`². The classifier predicts labels randomly, following a uniform distribution, with each class having equal probability.

²<https://scikit-learn.org/stable/modules/generated/sklearn.dummy.DummyClassifier.html>.

3.2.2 TF-IDF Logistic Regression Baseline

We train a text classification model using TF-IDF vectorisation³ with unigrams and bigrams (n-gram range 1-2), limiting the vocabulary to 150k features. The vectoriser is fitted on the training data split and then applied to the test data split with the learned vocabulary. We employ a multinomial logistic regression classifier with L2 regularisation supported by the LBFGS optimiser, with a regularisation strength of $C=50$. The model is trained until convergence or for a maximum of 100 iterations, with a random seed (16), and parallelised across 4 CPU cores. Model performance is evaluated using 5-fold stratified cross-validation with shuffling enabled to preserve class distributions.

3.2.3 DistilRoBERTa Baseline

We fine-tune a DistilRoBERTa⁴ Transformer model on the single-label classification task for 10 epochs, with the option for early stopping based on the validation F1-score. We use half-precision floating-point (fp16) for faster training and batch sizes of 16 for both training and evaluation. Since the main points of the articles are usually summarised at the beginning, we apply fixed truncation at the maximum limit of 512 tokens and discard any remaining text. The model is fine-tuned on the training dataset (with a fixed random seed of 16) and evaluated on the validation set after every epoch. Checkpoints and logs are saved at the end of each epoch, and the best model – based on the F1-score on the validation set – is selected and loaded at the end of training. We follow Spokorny et al. (2023)’s and Calamai et al. (2025)’s recommended parameter settings and train at a learning rate of $5e-5$, with 0.1 warmup ratio and 0.01 weight decay.

3.2.4 Zero-Shot Models

For the zero-shot experiments, we choose three open-source models of various sizes and architectures as zero-shot classifiers:

- Llama3.3-70B-Instruct⁵ by Meta
- Mistral-Large-2-Instruct-2411 (123B)⁶ by Mistral AI
- GPT-OSS-20B⁷ by OpenAI

All three models have a context window of 128k tokens. Llama3.3 and Mistral-Large are dense models fine-tuned for instruction-following, whereas GPT-OSS uses a mixture-of-experts architecture with 3.6B parameters per token activated to process inputs. We opt for different architectures to ensure that our findings are not limited to a specific model design or size, and to evaluate the benchmark’s consistency across multiple scales and resources. The specific choice of models is influenced by related literature to guarantee reproducibility, and by our task requirements: the three models are optimised for multilingual use and support longer, multi-step inputs and structured outputs; additionally, they are less prone to hallucination and have strong general reasoning abilities⁸.

³TFIDFVectorizer from Sklearn: https://scikit-learn.org/stable/modules/generated/sklearn.feature_extraction.text.TfidfVectorizer.html.

⁴<https://huggingface.co/distilbert/distilroberta-base>.

⁵<https://ollama.com/library/llama3.3:70b>.

⁶<https://ollama.com/library/mistral-large>.

⁷<https://openai.com/index/introducing-gpt-oss/>.

⁸All three models achieve an accuracy score above 84% on MMLU: <https://ollama.com/library/llama3.3:70b>, <https://mistral.ai/news/mistral-large-2407/>, <https://ollama.com/library/gpt-oss:20b>.

We use Ollama⁹ to run the models locally. In terms of time and compute cost, the models’ efficiency is proportionate to their size: GPT-OSS is the fastest and lightest model to run, and Mistral-Large 2 has the highest compute and memory requirements.

Since the SciDCC dataset does not contain descriptions or definitions of the labels, we create them using the Merriam-Webster¹⁰ and the Cambridge¹¹ Dictionary. We test several different prompt formats on small random subsets (50 examples) drawn from the training subset and select the formulations that produce the most coherent results. For the single-label classification task, we test prompts with and without label descriptions provided and report results in both setups. The label descriptions are not included in the prompts for the multi-label classification task. Our tests and findings regarding that design choice are described in more detail in Section 6.1 of the Appendix.

Finally, all models are set to generate deterministic outputs (temperature of 0.1, top_p of 1). We run preliminary experiments with different random seeds on a subset of 100 examples to assess output variability under deterministic constraints, and find that Llama3.3 and Mistral-Large 2 produce consistent outputs overall, with only one case each where outputs differ across runs. GPT-OSS shows a slightly higher variability of 7% across seeds, which we account for by running each experiment three times with seeds 16, 31, and 173, respectively, and report aggregated results.

Additionally, Llama3.2 3B¹² by Meta is included as a baseline for the multi-label classification and ranking experiments. Its settings follow those of the other LLMs: deterministic output with a temperature of 0.1, a top_p value of 1, and local access through Ollama.

3.2.5 Other Models

For the explorative study of the benchmark, we use several open-source and commercial LLMs via Ollama and other API services:

- Mistral-Medium-3.1 via Mistral AI Studio
- Mixtral-8x22B via Ollama (locally)
- Llama3.3-70B-Versatile via Groq
- GPT-OSS-120B via Ollama (locally)
- Qwen3.5-397B via Ollama Cloud

The models are selected to cover a broad range of characteristics, including different parameter sizes (from medium to very large models), architectures (dense and mixture-of-experts), and deployment settings (local inference and API services). The particular choice of model sizes is based on Lunardi et al. (2025)’s finding that larger models show a strong and significant positive correlation between accuracy and consistency, while smaller models tend to be consistent but incorrect in their predictions. We aim to conduct a comprehensive evaluation of the benchmark that is representative of diverse ecosystems and scales and captures a variety of reasoning and language understanding abilities. All models are set to deterministic outputs.

Furthermore, we access GPT-5 Nano via the OpenAI API to generate the paraphrased dataset variant. We follow OpenAI’s recommendations on prompt formatting and

⁹<https://ollama.com>.

¹⁰<https://www.merriam-webster.com/>, last consulted online on March 18th 2026.

¹¹<https://dictionary.cambridge.org/>, last consulted online on March 18th 2026.

¹²<https://ollama.com/library/llama3.2>.

format the input as a two-turn interaction, with a system instruction containing the paraphrasing prompt and a user message containing the article body. Our tests across different reasoning settings reveal that the minimal reasoning mode and medium verbosity level yield the highest-quality results. Paradoxically, a longer, more descriptive prompt leads the model to output the original article text unchanged, whereas a prompt with a direct command and a few added details produces the desired outcome. The prompt we use can be found in Section 6.1 of the Appendix.

3.3 Text Transformations

We perform a set of three separate lexical transformations on a word/sentence level: code-mixing, paraphrasing, and masking. All transformations aim to modify surface-level features of the text, such as keywords and phrases, while preserving the semantic meaning, structure, and formatting of the original article. Our goal is to assess whether or not models are susceptible to superficial variations that do not affect the underlying message of the text.

All transformations are screened systematically for errors and mismatches by randomly selecting 50 examples of each generated dataset and comparing them to the original texts. Limitations and peculiarities are reported in the respective subsection. Examples of the variants can be found in Section 6.2 of the Appendix.

3.3.1 Paraphrasing

Paraphrasing is a sentence-level transformation that modifies the surface form of the input without altering its semantic meaning. It includes both lexical and syntactic changes, resulting in variations of wording, word order, sentence boundaries, and overall text length. The transformation aims to assess whether models rely on pre-memorised textual patterns or can genuinely interpret and analyse the content of the text.

In our specific setup, the articles are rewritten in a colloquial, less scientific style with informal expressions and slang. We prompt (Section 6.1.1 of the Appendix) GPT-5 Nano to reformulate the articles ‘in Gen Z style’, as a more explanatory prompt with examples of the desired tone does not produce any satisfactory results. The manual analysis of the generated outputs reveals that the texts were indeed not paraphrased in Gen Z register but rather in an informal, pretending to be youth, slang from the early 2010s (‘yo, check this out’, ‘something wild happened’, ‘the fish were chilling’, etc.), despite the training cutoff for GPT-5 Nano being in 2024¹³. Some examples of the modifications include ‘chill’, which is used 4 times in the original dataset in the sense of cooling off and as part of the phrase ‘wind chill’; in the paraphrased dataset, it is used 509 times, in various phrases and as different parts of speech. The use of the adjective ‘wild’ is almost doubled: from 1434 times in the original SciDCC to 2460 times in the transformed dataset. Lastly, the word ‘vibe’ is found only once in the original texts as part of a humorous title of a study, and in the paraphrased texts, it is used 5315 times.

Despite the style inconsistencies with the prompt, the contents of the articles are, to a large extent, preserved correctly, and no relevant information is lost or modified in a way that would deem it incomprehensible. When compared to the original sentences, some paraphrased ones may appear as if key terms were replaced with nonsensical phrases, which may disturb the overall understanding of the text. For instance, the sentence ‘When it comes to air quality, changing climate patterns fuel wildfires

¹³<https://developers.openai.com/api/docs/models/gpt-5-nano>.

and their dangerous smoke.’ is paraphrased to ‘Basically, climate vibes are fueling wildfires and smoky air.’ Still, the surrounding context makes it clear that ‘climate vibes’ refers to climate change, and while that does not erase the odd phrasing, the core semantic meaning of the text is successfully conveyed. Such cases appear in almost all paraphrased articles, but due to how context- and phrase-dependent they are, semantic errors are difficult to quantify. Lastly, all articles appear shorter than the originals, as lengthy scientific terms are replaced with more straightforward phrases, and longer sentences from the original text are often rewritten into multiple shorter ones.

Despite the prompt instructions, 7 out of the 50 reviewed articles contained meta-comments in the beginning or at the end of the paraphrased text (‘The user has requested...’, ‘Here is the paraphrased article:..’, ‘I paraphrased the article...’, ‘Would you like me to provide ...’, ‘Would you prefer...’). Since we used structured JSON output as a default output format, 5 of the articles also contained output markers and nonsensical strings: ‘}}’, ‘]]end of structured output’, ‘]]end of JSON output]]][[’. Since the meta characters and comments appeared at random and in different sequences, an automatic cleaning procedure was not 100% effective, so we inspected the entire paraphrased dataset manually. In particular, we searched through the texts for typical keywords like ‘output’, ‘JSON’, ‘would you’, ‘do you’, ‘paraphrased’, ‘paraphrases’, ‘dataset’, ‘here is’, ‘there is’, ‘I have’, etc., and characters like ‘}’, ‘]’, as well as combinations thereof. To our knowledge, the final paraphrased dataset is clean of any model outputs different from the articles themselves.

3.3.2 Keyword Masking

Keyword masking is a word-level transformation that includes replacing selected content words and phrases with a mask token (<MASK>), while preserving the overall sentence structure. It results in partial information loss, which increases task difficulty, as models must derive the article’s topic from the remaining unmasked context. This variant of the dataset is used to assess model robustness to missing keywords and phrases.

The keyword masking pipeline operates by first splitting each article into sentences using the NLTK (Loper and Bird, 2002) sentence tokeniser¹⁴; sentences are then further chunked into tokens. Next, keywords are extracted from all chunks with KeyBERT¹⁵ on top of all-MiniLM-L6-v2 Sentence Transformer¹⁶ by comparing each token’s embedding with the article’s embeddings. If the token’s embedding is similar enough to the article’s embedding, it is added to a list of keyword candidates. The extracted keywords are then aggregated at a sentence level. For each sentence, keywords are ranked based on their embedding similarity scores, and a masking fraction variable is applied to control the fraction of ranked keywords that are to be masked (for instance, a 25% ratio would mean that the top-25% ranked keyword candidates in each sentence are selected for masking). There is also an upper keyword limit based on word count to ensure a balanced degree of masking across sentences of varying lengths. Finally, the selected keywords are masked in the original text, and the pipeline returns the masked article. We also set some additional parameters, such as overlapping tokens (20) to preserve context across sentences and a minimum number of keywords to mask (1). In reality, due to the token overlap, the algorithm masks more words per sentence than the set maximum, which is why several experiments with different parameters were needed to find a good balance between quantity and quality of masking.

¹⁴https://www.nltk.org/api/nltk.tokenize.sent_tokenize.html.

¹⁵<https://maartengr.github.io/KeyBERT>.

¹⁶<https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2>.

Our initial intent was to use an LLM for this transformation. However, the preliminary experiments with multiple open-source and commercial LLMs showed that, despite its limitations, our algorithm produces more coherent variants than an LLM, which, in turn, selects keywords at random, clusters the masked keywords in one paragraph, leaving the rest of the text unmasked, or often even fails to process the entire article.

We create three masked datasets with different masking degrees: an easy (25%), a moderate (50%), and a difficult (75%) variant. The manual inspection of a 50-article sample of each of the datasets confirms that in all articles, only relevant content keywords and phrases are masked, and the structure, length, and boundaries of the sentences are fully preserved. The masked keywords are evenly distributed across the entire articles, and depending on the masking degree, every sentence contains 2-4 masked keywords or phrases. In about 5-6 articles per variant, certain sentences do not contain any masked keywords – probably because the algorithm did not detect any candidate keywords that were similar enough to the overall text’s subject. At the highest masking degree, the reading flow is obstructed, but all texts are understandable despite the increased number of masked tokens. Due to time and computational constraints, in the multi-label classification and ranking experiments, we evaluate models on a randomly drawn subset of 1000 articles for each masked variant (seed 16) rather than the full masked datasets.

3.3.3 Code-Mixing

Code-mixing is a word-level transformation that introduces lexical substitutions from a second language into the original text, while preserving the sentence structure and, to a large extent, the grammar of the original input. The goal of the transformation is to evaluate whether models remain robust to cross-lingual lexical perturbations or rely on monolingual surface patterns.

Our setup performs synthetic code-mixing on the original English articles by systematically replacing content keywords and phrases with their German translations. Similar to the masking pipeline, the algorithm splits each text into sentences, then further into token-sized chunks, and extracts relevant candidate n-grams using their embedding similarity scores. We use the paraphrase-multilingual-MiniLM-L12-v2 Sentence Transformer¹⁷ because of a post-processing step that requires multilingual knowledge. Next, the extracted keywords are ranked and aggregated back on a sentence level. A limited number of high-scoring, non-overlapping phrases per sentence are translated into German using the neural machine translation model opus-mt-en-de (Tiedemann and Thottingal, 2020). Following that step, the translations are analysed using the `de_core_news_sm` spaCy model¹⁸ for German part-of-speech tagging to capitalise German nouns. A post-processing function is implemented to prevent the same translation from being repeated (if a translation is already covered by a longer translated phrase in the same sentence), to block parts of the sentence from being modified more than once, to segment tokens that were merged accidentally (for cases like ‘theearth’ -> ‘the Earth’), and to remove nonsensical strings of repeated tokens (such as ‘sssssss’, ‘<><><><>’ or ‘clasclasclas’). The result is a semantically consistent, code-mixed text with controlled lexical perturbations.

Furthermore, we set additional parameters, such as the n-gram range (2,3) of translated keywords – necessary due to how German nouns are constructed, a minimum embedding similarity score threshold of 0.3 for a keyword to be considered relevant for translation (we find that lower values introduce more noise into the

¹⁷<https://huggingface.co/sentence-transformers/paraphrase-multilingual-MiniLM-L12-v2>.

¹⁸<https://spacy.io/models/de>.

translated phrases), and a context overlap of 20 tokens between sentences to preserve context.

The manual inspection of the generated code-mixed articles reveals phrase repetitions that were not fully removed by the post-processing function. Therefore, we implement an additional cleaning step for the remaining repetitive strings, which further reduces their occurrence. Additionally, we refine the prompt in the zero-shot experiments with code-mixing to ignore repetitive longer phrases and sentences, should they occur more than once. In terms of structure, the transformed articles have the same length as the original ones, and sentence boundaries are well preserved. All articles contain at least one sentence with errors related to the main verb’s placement, as translated verbs are plugged back into the same positions they would typically hold in an English sentence, while in German, the verb is written at the end of the sentence. Capitalisation of the German nouns is mostly correct with a few exceptions when a noun was likely not detected as such by the capitalisation function (in about 8 cases throughout the 50 articles). In 11 cases, verbs are incorrectly translated as nouns or vice versa, leading to some inconsistencies in the text structure. There are very few mistranslations (7 cases in total), largely due to homonyms in English. Despite the listed issues, the text is coherent and understandable.

3.4 Annotation Procedure

The following section outlines the annotation procedure used to create a manually labelled subset of 60 articles, which serves as a reference to validate model performance on the multi-label classification and ranking tasks. The subset for annotation is randomly sampled from the original SciDCC dataset using the same random seed (16) as in the perturbation experiments, and therefore covers all transformation variants. We apply the EffiARA (Efficient Annotator Reliability Assessment) annotation framework (Cook et al., 2025) to maximise the number of annotated samples reliably with a limited annotation budget. We use it to distribute samples, compile annotations, and calculate inter- and intra-annotator agreement.

3.4.1 Guidelines Creation and Refinement

We work iteratively and conduct two preliminary rounds of annotations to adapt the task and the guidelines before choosing a final version. In the first round, annotators were asked to read the article and select one main label and up to three secondary labels, providing the reasons for that choice in a short sentence. However, this structure did not account for cases where more than 4 labels would be suitable, or for varying degrees of relevance among the secondary labels. Despite that, the written responses contributed to a better understanding of the edge cases in the data. In the preliminary second round, we adopted a point distribution method, where annotators were allowed to select up to 10 labels and had a budget of 10 points in total to distribute among them. The limit on labels was aimed at avoiding floating numbers and too big evaluation granularity. While the annotators mostly agreed on the sets of selected labels, the point allocation method led to a very low inter-annotator agreement, with some annotators reporting that decisions about exact point budgets across labels were somewhat arbitrary and at random. Another downside of the method was that models generally tend to hallucinate when prompted to assess their own performance and could not assign budgets the same way people do (besides inconsistent responses, models often distributed 9 or even 12 points in total).

Besides the task design, we also experimented with the label definitions and tested to see if including them in the guidelines had any noticeable impact on the inter-annotator

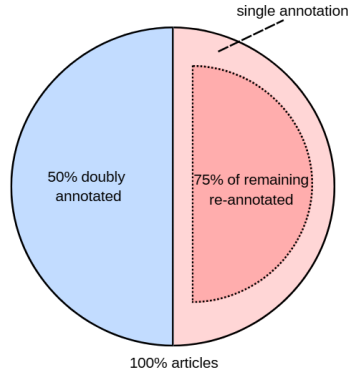


Figure 3.3: Distribution of the annotated samples. Half of the articles are annotated by two different annotators, and approximately 75% of the remaining articles are re-annotated by the same annotator (double proportion = $1/2$, re-proportion = $3/4$).

agreement. Unlike model performance (experiments described in more detail in the Appendix, Section 6.1.3), the presence of label descriptions substantially improved the consistency of the human annotators, both in terms of responses across different annotators and responses from the same annotator. Annotators reported that the label descriptions were especially useful in edge cases and improved confidence in their annotations.

The final version of the guidelines includes two tasks: multi-label classification and ranking according to the labels’ relative importance to the text. There is no limit to the number of labels that may be selected. The second task includes a subtask, where annotators are asked to provide a score on how confident they are in their assessment. The full guidelines can be read in Section 6.3 of the Appendix.

3.4.2 Annotation Framework

The EffiARA framework is designed for multiclass multi-label classification tasks with limited resources and supports the annotation process at multiple levels. Given the scientific style of the articles, we work with 5 annotators with near-native English proficiency and experience in reading and interpreting specialised academic literature. Using EffiARA’s distribution algorithm, we were able to estimate in advance the cost of the annotation project and to allocate the articles to the annotators. Since having all annotators label all articles is too costly, we use the framework to create a sample distribution that provides reliable estimations of the inter- and intra-annotator agreement through partial overlap of the annotated articles (Figure 3.3). In total, the annotated subset consists of 60 randomly selected articles, of which 30 are doubly annotated, and 20 are re-annotated.¹⁹ We use the confidence scores provided by the annotators to identify particularly difficult articles.

To reduce the risk of annotation bias and non-independent annotations, we split the articles into two rounds with a time period of several days in between. In the first round, the annotators receive about 80% of the subset assigned to them, and in the second round, they are given the articles that require re-annotation.

¹⁹The framework targets a re-annotation proportion of approximately 75% of the remaining singly annotated articles; however, the exact number depends on the total number of articles and annotators. In this dataset, this corresponds to 20 out of 30 articles.

3.4.3 Postprocessing and Evaluation

We aggregate the collected annotations in a dataframe object with a column for each annotation from each annotator. Next, we calculate intra- and inter-annotator agreement through the EffiARA evaluation framework: double annotations are compared across annotators and contribute to the overall inter-annotator agreement score, and re-annotations measure intra-annotator agreement. Since the default setting of the EffiARA annotator reliability estimator assumes single-label inputs, we implement a custom multi-label confidence label generator method to fit our annotated data. Instead of treating each annotation as a different single-label string ($A1 = \text{"Biology, Animals, Zoology"}$), the custom method converts labels into sets ($A1 = \text{"Biology", "Animals", "Zoology"}$) or preserves them as ranked lists. Another customisation includes the metric used to estimate inter- and intra-annotator agreement. The EffiARA framework offers built-in metrics like Krippendorff’s alpha and Cohen’s kappa, but they do not support partial agreement and ranked labels. Therefore, we adapt the framework by replacing these metrics with Jaccard similarity for unordered multi-label sets (multi-label classification task) and rank-biased overlap (RBO) for ranked label lists (multi-label ranking task). This way, the calculated agreement reflects both partial overlap and label ordering, while preserving EffiARA’s original reliability estimation framework.

3.5 Evaluation

The evaluation methodology of this study consists of multiple parts, including single- and multi-label classification evaluation, model behaviour analysis, benchmark robustness and consistency testing, as well as comparison between model predictions and human judgment.

In the exploration phase, we evaluate several models on the original single-label classification task and report performance on the test split using macro-averaged F1-score as the main evaluation metric. We opt for macro-averaging because of the highly imbalanced class distribution of the SciDCC dataset. The method forces the classifiers to distinguish between the classes rather than rely on their distribution and ensures the comparability of our results with prior work. Besides F1-scores, we use accuracy to measure individual LLM performance and flag inconsistent labels in the dataset.

In the second part of the study, we aim to reduce the benchmark’s dependence on single labels being the ‘gold’ standard and shift the evaluation paradigm from single-label correctness as an evaluation metric to methods from the adversarial domain, such as robustness, consistency, and relative performance across transformations. We formulate two new tasks: a multi-label classification task and a multi-label ranking task.

Robustness and consistency of multi-label classification are evaluated through the Jaccard similarity index, which measures similarity between two asymmetric binary vectors, i.e. two sets of labels. It is calculated as the intersection of two sets divided by their union. The index ranges between 0 and 1, where 1 is a total overlap of the label sets and 0 means that the two sets do not share any common members. We compute the Jaccard similarity between all pairs of transformations across four zero-shot LLMs and examine whether predictions remain stable under transformations (robustness) and whether different variants produce identical predictions (consistency). Additionally, we use precision and recall to evaluate the effect of the transformations on predictions’ stability between the original, non-transformed articles and the perturbed variants. Recall measures how many labels derived from the clean articles are preserved in each

variant after transformation; high recall means that most original labels are preserved. Precision quantifies the extent to which transformations introduce additional labels, and high precision means that transformations add to the overall label count.

We evaluate the multi-label ranking task using the rank-biased overlap (RBO) score (Webber et al., 2010), which supports non-conjoint ordered lists of various lengths, i.e. items that appear in one of the lists do not necessarily appear in the other. The RBO is an intersection-based similarity measure that calculates agreement based on overlapping items between the two lists and their relative rank positions; top-ranked items are assigned larger weights, and (dis)agreement at the top of the lists is penalised/awarded more than (dis)agreement towards the bottom. The RBO is bounded between 0 and 1, where 0 indicates no observed agreement, and 1 refers to the lists being completely identical. The RBO uses a tunable parameter ρ , which determines the contribution of the top ranks to the final RBO score; we apply a ρ value of 0.9, so that the top items have the largest impact, but the lower ones contribute a small amount, too. One key limitation of the RBO score is that it does not penalise differences in the length of the compared lists; for this reason, in addition to the RBO, we apply an adjusted metric we call the RBOx score (3.1), defined as the extrapolated RBO score²⁰ between two ranked lists A and B, multiplied by the ratio of the shorter list length to the longer list length, raised to the power 0.2. The RBOx accounts for length mismatches by softly penalising unequally long lists, while also preserving the overlap agreement when the mismatches are negligible.

$$\text{RBOx}(A, B) = \text{RBO}_{\text{ext}}(A, B) \left(\frac{\min(|A|, |B|)}{\max(|A|, |B|)} \right)^{0.2} \quad (3.1)$$

²⁰<https://github.com/changyaochen/rbo>.

4 Results

The following chapter presents the results from our experiments. In the first section, we introduce the outcomes of the benchmark exploration. Next, we report on models’ consistency, robustness, and relative performance, and evaluate the inter-model agreement. In the last section, we share our findings regarding the annotation procedure and estimated inter- and intra-annotator agreement. Finally, we assess the models’ behaviour and performance against the human annotations.

4.1 Benchmark Exploration

This section summarises the results from the SciDCC benchmark exploration phase. We evaluate the performance of three baselines and three LLMs on a single-label text classification task by article body. The baselines include a random classifier, a TF-IDF logistic regression model, and a fine-tuned Transformer model; the three LLMs are tested in a zero-shot setting, with and without label descriptions in the prompt.

Table 4.1 presents the macro F1-scores of all models on the test set, as well as the results from the zero-shot experiments on the entire dataset. The random baseline achieves an F1-score of 1.44%, which is expected given the high number of labels (20) and the class imbalance of the dataset. The TF-IDF model performs substantially better, reaching an F1-score of 49.57%. The fine-tuned DistilRoBERTa model has the best performance overall, with a score of 50.04%. The LLMs in a zero-shot setting perform notably worse than the supervised classifiers: Without label definitions, Mistral-Large 2 achieves an F1-score of 33.77%, Llama3.3 70B 24.62%, and GPT-OSS 20B 34.36%. Providing label definitions in the prompt improves performance for Mistral-Large 2 (+9.2%) and Llama3.3 70B (+4.4%), whereas GPT-OSS 20B does not benefit from them. Despite the improvements, the performance of the zero-shot methods remains below that of the supervised baselines. The results of the zero-shot LLMs on the full dataset without label descriptions indicate a pattern similar to the results without definitions on the test set, with GPT-OSS 20B achieving the highest zero-shot performance (34.89%).

Overall, the results align with prior work. Similar to Calamai et al., 2025, whose evaluation is conducted on article summaries, the TF-IDF logistic regression model and the fine-tuned DistilRoBERTa classifier achieve nearly identical F1-scores and outperform the zero-shot models and the random baseline. The LLM performance in a zero-shot setting without label descriptions is also comparable to the scores reported

Dataset split	Random	TF-IDF	DistilRoBERTa	Mistral-Large 2	Llama3.3 70B	GPT-OSS 20B
Test set (no definitions)	1.44	49.57	50.04	33.77	24.62	34.36
Test set (with definitions)	-	-	-	42.96	28.99	34.03
Full dataset (no definitions)	-	-	-	28.89	22.65	34.89

Table 4.1: Macro F1-scores (%) of baselines and zero-shot experiments on the SciDCC dataset for single-label classification by article body.

by Kurfali et al., 2025, who use a wider range of inputs (combinations of article body, title, and summary).

We analyse the errors of the evaluated models, comparing their most frequent mispredictions. For both the TF-IDF and the fine-tuned DistilRoBERTa models, the most commonly confused true classes are Climate, Biology, Animals, and Geography, and the most over-predicted classes are Endangered Animals, Weather, Biology, and Agriculture & Food. These confusion patterns are reflected in the TF-IDF feature weights: Several highly weighted terms appear across multiple categories, making the distinction between them difficult. For example, terms like ‘cell(s)’ and ‘protein(s)’ are among the top 3 most frequent terms both in the Biology and Biotechnology classes, and broader terms like ‘climate’, ‘water’, and ‘solar’ appear equally often in Environment, Climate, Weather, and Geography. This lexical overlap shifts the class boundaries and offers an explanation for why said classes are more frequently confused by the TF-IDF model than other classes with stricter lexical registers, such as Hurricanes & Cyclones and Microbes.

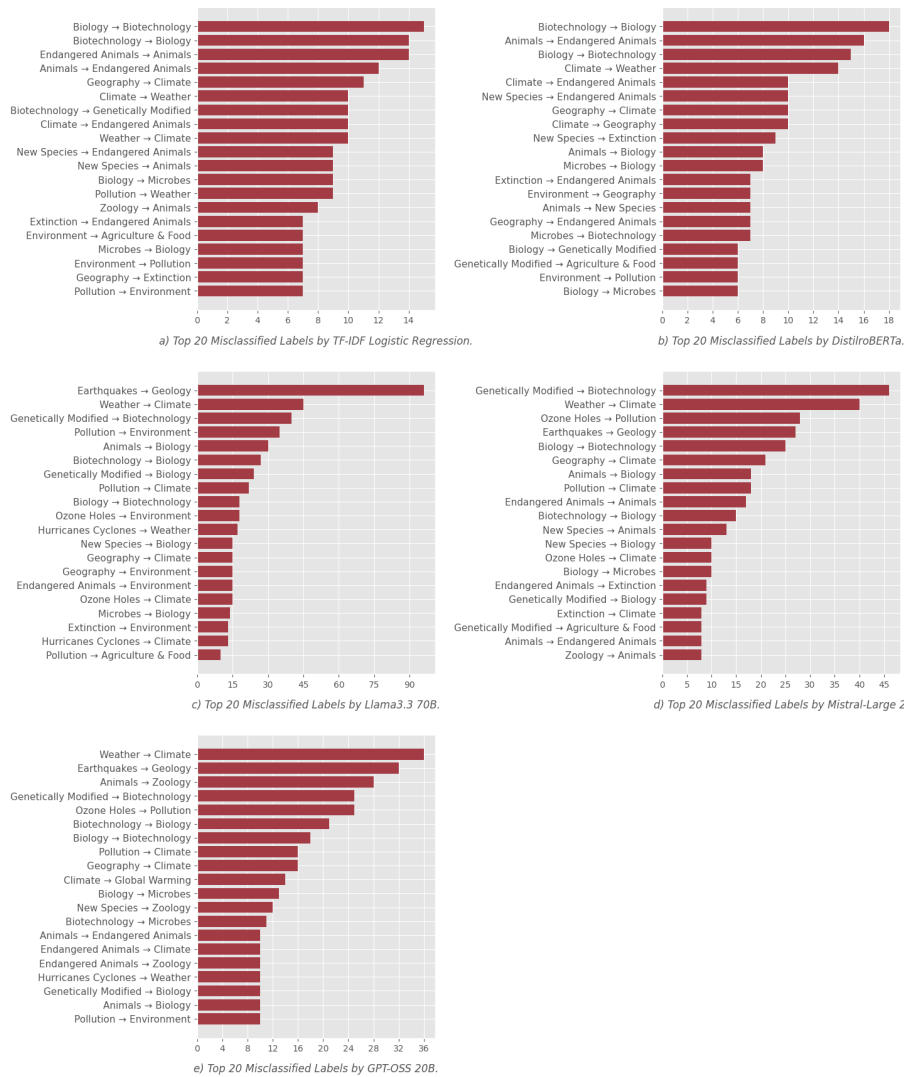


Figure 4.1: Top 20 misclassified labels per baseline/method. The y-axis represents the number of occurrences. The arrows point from the original labels towards the predicted.

Figure 4.1 shows the most frequently misclassified labels for each of the evaluated

models. While the exact numbers of cases vary across models, all methods exhibit class bias towards broader topics with more general, all-encompassing keywords and umbrella terms (Climate, Biology, Biotechnology). Especially the LLMs collapse fine-grained distinct categories into wider ones (for instance, the classes Weather, Pollution, Geography, and Ozone Holes are collapsed into the category Climate, and the classes Animals, Endangered Animals, New Species, and Genetically Modified are merged into the labels Zoology or Biology). Finally, some classes tend to be confused in pairs, such as Biology and Biotechnology, Animals and Endangered Animals, and Geology and Earthquakes, which further indicates that misclassifications are hardly random but rather systematic due to strong semantic and lexical overlap between hierarchically or thematically related classes.

LLM Error Analysis To better understand LLM behaviour and error types, we conduct an in-depth analysis of a subset of 100 randomly sampled articles, comparing the performance of seven LLMs. We consider the agreement of at least four of the seven models on the same label a majority.

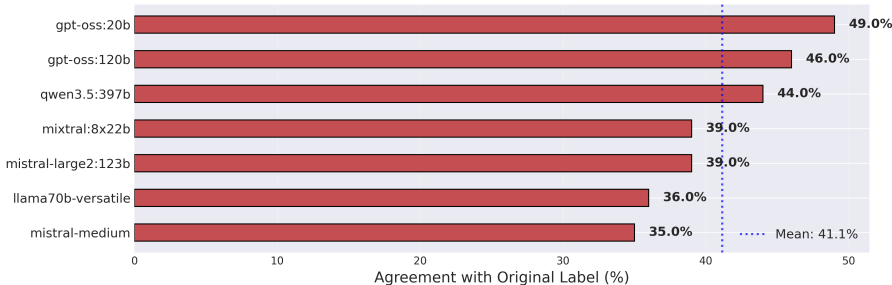


Figure 4.2: Individual LLM performance on a random subset of 100 articles (accuracy scores). The average agreement with the original label is 41.1%.

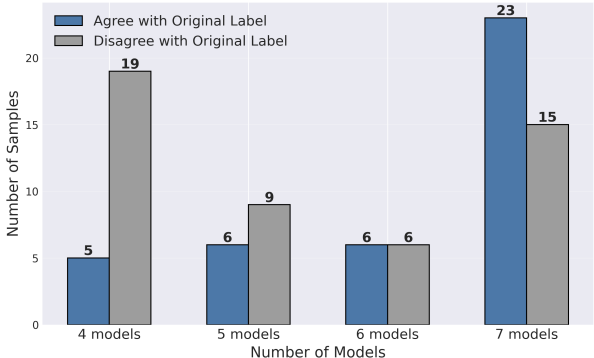


Figure 4.3: Majority size distribution (when 4+ models agree).

Figure 4.2 shows the individual performance of each of the evaluated models on the subset. The average agreement with the original label is 41.1%, with GPT-OSS 20B having the highest score (49%) overall and Mistral-Medium having the lowest (35%). The results also reveal major disagreement between the models’ predictions and the original dataset labels: In 49% of cases, the majority of models assign a different label than the original one, and only 40% of samples show majority agreement with the original label (Figure 4.3). Despite this, inter-model agreement remains relatively high (66.6%), indicating that models tend to agree with each other more than with the original label. The most agreeing pair of models is GPT-OSS 20B and GPT-OSS 120B

Error Type	Count	Example
Multiple Labels Possible	21	Genetically Modified →Biotechnology
Interchangeable Labels	10	Earthquakes →Geology
Wrong Label	14	Ozone Holes →Climate
Borderline Wrong Label	4	Animals →Endangered Animals

Table 4.2: Summary of errors in disagreement cases. The arrow points from the original label toward the predicted label.

(in 81% of the cases), and the least agreeing pair is Qwen3.5 and Mistral-Large 2 (in 54% of agreed-on samples).

We manually analyse the 49 disagreement cases and define four error types, explaining the mismatch between the majority vote and the original label:

- **Multiple Labels Possible:** the original label is one of several different labels that are applicable.
- **Interchangeable Labels:** the original label is too closely related to (an)other label(s).
- **Wrong Label:** the original label does not apply in the specified case.
- **Borderline Wrong Label:** technically, the original label applies, but is considerably less relevant than (an)other label(s).

The results are summarised in Table 4.2. The majority of disagreement cases fall into the category *Multiple Labels Possible* (21 cases) – a finding that highlights the ambiguity of the articles’ contents. Furthermore, the relatively high number of incorrect labels (14 cases) suggests that a large portion of the dataset’s classes may be assigned inaccurately or inconsistently. The category *Interchangeable Labels* accounts for 10 cases and indicates an additional weakness of the benchmark, namely label redundancy. Combined with the inherent complexity of the texts, this design mismatch reinforces the patterns observed in the earlier error analysis, particularly the overlap between semantically or hierarchically connected categories and the merging of categories, resulting in granularity loss.

4.2 Multi-Label Classification and Ranking

We assess the reliability of the redesigned benchmark setup by analysing the consistency and robustness of model predictions across three text transformations. We define and evaluate two separate tasks: multi-label classification and ranking.

Label Selection Table 4.3 reports the multi-label classification results across transformation pairs in combination with the unperturbed original texts using the Jaccard similarity index, precision, and recall. For the calculation of precision and recall, the label sets predicted from the clean articles are taken as the base against which predictions from the transformed articles are compared. Overall, multi-label classification shows great stability under lexical perturbations like code-mixing and paraphrasing, and very good stability under information loss through masking across all models.

Unsurprisingly, the larger models exhibit stronger label overlap than the smaller model, and maintain consistent performance under perturbations. Mistral-Large 2 and Llama3.3 70B demonstrate the strongest robustness, with high Jaccard scores (0.83–0.90) all across and corresponding high precision (0.90–0.96) and recall (0.90–0.94), which indicate strong agreement between the label sets derived from the clean

Variant Pair	Metric	Mistral-Large 2	Llama3.3 70B	GPT-OSS 20B	Llama3.2 3B**
Original – Code-Mixed	Jaccard	0.894	0.889	0.839	0.703
	Precision	0.948	0.967	0.916	0.828
	Recall	0.933	0.915	0.898	0.805
Original – Paraphrased	Jaccard	0.894	0.902	0.834	0.709
	Precision	0.945	0.954	0.904	0.858
	Recall	0.934	0.942	0.904	0.787
Original – Masked 0.25*	Jaccard	0.853	0.845	0.852	0.624
	Precision	0.930	0.910	0.919	0.759
	Recall	0.899	0.919	0.920	0.769
Original – Masked 0.5*	Jaccard	0.807	0.824	0.816	0.593
	Precision	0.910	0.897	0.898	0.702
	Recall	0.863	0.908	0.901	0.786
Original – Masked 0.75*	Jaccard	0.771	0.812	0.788	0.560
	Precision	0.888	0.887	0.876	0.661
	Recall	0.835	0.901	0.880	0.778

Table 4.3: Robustness and consistency of model predictions under text transformations in the multi-label classification task measured via Jaccard similarity index, precision, and recall. *Masking applied to a 1000-article subset; all other transformations apply to the full dataset. **Model Llama3.2 3B only tested on subsets of 1000 articles.

and the transformed texts. GPT-OSS 20B exhibits slightly lower robustness under the same transformations, although its behaviour remains comparatively stable, with well-balanced precision and recall. In contrast, the smaller Llama3.2 3B model shows lower robustness across all transformations, with Jaccard scores falling to 0.703 and 0.709 for code-mixed and paraphrased inputs, respectively. This suggests that the smaller model relies more heavily on surface-level textual cues when identifying relevant labels and is less capable of preserving predictions consistently.

As the masking scale increases in the masked transformations, all models note a decrease in output consistency. Mistral-Large 2 has the steepest decline among larger models, dropping from a Jaccard score of 0.853 at 0.25 masking to 0.771 at 0.75 masking. In comparison, Llama3.3 70B is the most resilient, as it maintains a comparatively high Jaccard score of 0.812 even at the most aggressive masking level. Its recall remains particularly strong (0.901), which indicates that it continues to choose most of the originally selected labels despite substantial information loss. GPT-OSS 20B follows a similar, though slightly weaker pattern, while Llama3.2 3B has the largest relative degradation, falling to a Jaccard score of 0.560. The results also show that Llama3.2 3B has the most shifts in sets’ structure, adding and removing labels across transformations.

We further test the benchmark’s internal consistency by comparing pairwise label-set similarity across all variants using the Jaccard index. We specifically focus on the similarity between transformed variants (e.g. Code-Mixed–Paraphrased and Masked variant pairs), which remains consistently high across the three larger models and consistently lower for the smaller one. For Mistral-Large 2, Llama3.3 70B, and GPT-OSS 20B, the similarity scores between predictions from the code-mixed and the paraphrased variants (0.87–0.88) suggest that the produced label sets share a good portion of members despite the differing text modifications. Llama3.2 3B records a lower score of 0.67, showing greater sensitivity to change in input formulation, and the score is aligned with the weaker robustness observed across the original–transformed pairs in Table 4.3.

The predictions across the three masking levels also show relatively strong agreement for the stronger models, with 0.83–0.89 agreement at 0.25 and 0.5 masking and 0.82–0.89 at 0.5–0.75 masking. Llama3.2 3B lies at 0.67 Jaccard when 25–50% of keywords are

masked and at 0.68 when 50-75% are masked, indicating that while increased masking reduces context, the larger models continue to produce comparatively strong label sets. In contrast, the smaller model is more susceptible to information loss. Furthermore, lower similarities are observed between more distant masking scales, particularly between the 0.25 and the 0.75 masking levels, where scores decrease to 0.70-0.77 across Mistral-Large 2, Llama3.3 70B, and GPT-OSS 20B, and to 0.59 for Llama3.2 3B. Despite that decline and inter-model differences, all models retain above-average robustness across transformed-variant pairs, and the overall patterns show strong stability.

Ranking The results from the ranking task are presented in Table 4.4. We use two rank similarity metrics: the rank-biased overlap (RBO) and an adjusted RBO (RBOx) that penalises set length mismatches. Similarly to the multi-label classification task, we report similarity scores between variant pairs.

For all models, ranking stability remains consistently high under code-mixed and paraphrased inputs. RBO scores are close to 0.96-0.97 for Mistral-Large 2 and Llama3.3 70B, meaning that the relative ordering of predicted labels is mostly preserved despite transformations. Across all variant pairs, the corresponding RBOx scores are slightly lower than the standard RBO (by 0.01-0.02). We can therefore conclude that degradation is not only caused by changes in rank ordering but also by differences in the length of the prediction lists, as also observed in the first task, which is especially evident in the case of Llama3.2 3B, as it has the lowest precision, recall, RBO, and RBOx scores overall.

Overall, the masked variants yield lower scores than the lexical transformations, and the noise introduced through masking results in a consistent, gradual decrease in ranking similarity as the masking scale increases. For instance, as the masking level increases, Mistral-Large 2 drops from 0.91 (RBOx) at 0.25 masking to 0.86 at 0.75 masking. Llama3.2 3B shows a similar decline from 0.78 to 0.72. GPT-OSS 20B and Llama3.3 70B follow a slightly different pattern, as they maintain relatively strong robustness at 0.25 and 0.50 masking levels but decline more in the highest masking setting. Still, even under 0.75 masking, models retain reasonably high similarity scores.

Among the evaluated LLMs, Llama3.3 70B shows the strongest overall robustness, achieving the highest scores across most transformation settings. Mistral-Large 2 has comparable stability, with slightly greater degradation under heavier masking, whereas GPT-OSS 20B and, particularly, Llama3.2 3B exhibit greater sensitivity to transformations, despite maintaining strong ranking consistency overall. All in all, ranking performance remains stable across transformations: stronger models consistently perform at a high level, while the weaker model consistently performs at a lower level.

Inter-Model Agreement Analysis Although models produce stable classification and ranking results overall, we further explore inter-model agreement to test whether the LLMs converge on the same predicted label sets or each model is persistent in its own predictions. The results from that exploration are presented in Figure 4.4.

There is substantial overlap in multi-label classification (about 0.5-0.7 Jaccard on average), which indicates that the models often choose similar labels. However, that agreement is not uniform: GPT-OSS appears most aligned with the other models, whereas Llama3.2 shows the lowest agreement. Out of all models, GPT-OSS has the highest agreement with both Mistral-Large 2 and Llama3.3, while the two models share a moderate overlap with each other. There is, however, no distinct pairwise alignment among any of the LLMs. It is also notable that the two Llama models exhibit higher agreement with each other than Llama3.2 does with other non-Llama models.

Variant Pair	Metric	Mistral-Large 2	Llama3.3 70B	GPT-OSS 20B	Llama3.2 3B**
Original – Code-Mixed	RBO	0.962	0.972	0.939	0.865
	RBOx	0.946	0.952	0.914	0.835
Original – Paraphrased	RBO	0.959	0.974	0.938	0.878
	RBOx	0.943	0.956	0.912	0.845
Original – Masked 0.25*	RBO	0.917	0.945	0.958	0.822
	RBOx	0.913	0.923	0.931	0.782
Original – Masked 0.5*	RBO	0.913	0.937	0.944	0.799
	RBOx	0.887	0.911	0.911	0.756
Original – Masked 0.75*	RBO	0.889	0.929	0.927	0.770
	RBOx	0.860	0.902	0.891	0.725

Table 4.4: Robustness and consistency of model predictions under text transformations in the multi-label ranking task measured via RBO and RBOx. *Masking applied to a 1000-article subset; all other transformations apply to the full dataset. **Model Llama3.2 3B only tested on subsets of 1000 articles.

This is likely due to shared architectures, training, and tuning within the Llama family, which may lead to more similar prediction behaviour despite the size difference.

In contrast, models are far less aligned in their ranking predictions. Even though they seemingly select similar labels, the evaluated LLMs disagree on relative label relevance. Llama3.3 has the highest overall agreement with other models – 0.29 RBO with both Mistral-Large 2 and GPT-OSS, while Llama3.2 consistently shows the lowest alignment with non-Llama models (0.21 RBO).

The cross-model agreement analysis reveals that even though models appear highly consistent in their own ranking behaviour, compared with other models, their ranking performance is hardly robust. The low inter-model agreement suggests that each model might be following its own ranking logic, failing to converge towards a more stable, gold-standard-resembling prediction.

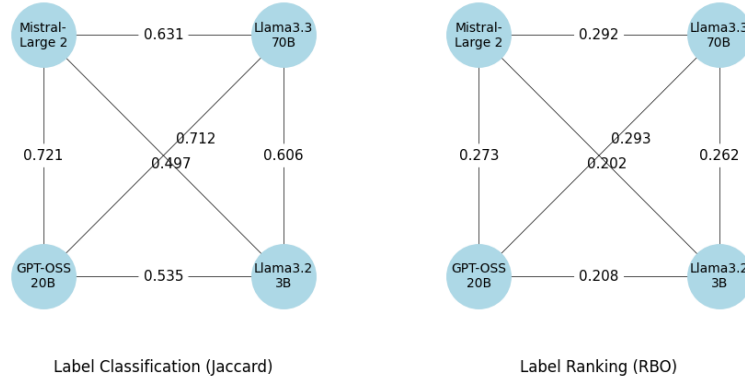


Figure 4.4: Inter-model agreement on multi-label classification and ranking. Scores are calculated on a 1000-article subset for Llama3.2 3B and on the entire datasets for the other models.

4.3 Annotation Results

The following section lays out the results from the annotation procedure and compares them to the observed model performance. We report results across five annotators and evaluate the annotated samples through the EffiARA framework according to the method described in section 3.4.

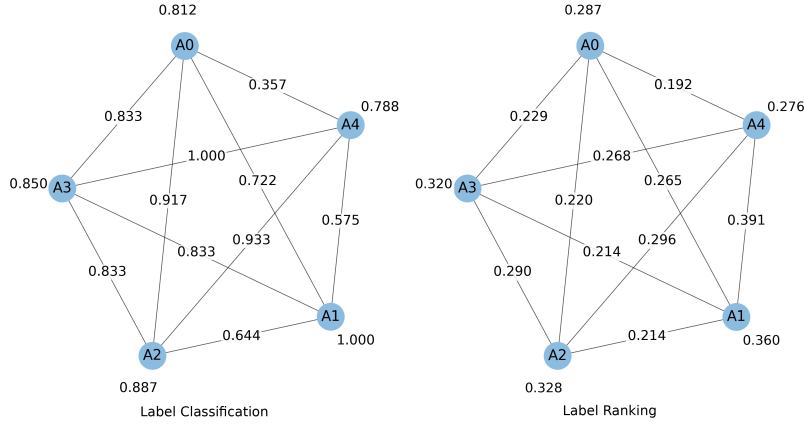


Figure 4.5: Recorded inter- and intra-annotator agreement from the annotation procedure within the EffiARA framework. Node values present annotators’ self-consistency (i.a., intra-agreement scores), calculated by comparing samples with their respective re-annotations by the same annotator; edges refer to the estimated pairwise inter-annotator agreement, derived from double annotations of the same sample. All annotators have annotated and re-annotated an identical number of articles.

Annotator Agreement We present the inter- and intra-annotator agreement across the two multi-label tasks in Figure 4.5. Multi-label classification uses the Jaccard similarity index for pairwise agreement, and multi-label ranking is calculated based on the RBO ($\rho=0.9$) between the annotated pairs of samples. For multi-label classification, intra-annotator agreement is consistently high across all annotators, which suggests that the annotators were mostly certain in their assessment when re-evaluating the task. Although a small number of annotator pairs exhibit lower agreement scores (e.g., A0–A4), the overall reliability of the task remains high, as annotators in lower-agreement pairs still achieve strong agreement with multiple other annotators. Disagreement is, therefore, limited to specific pairings and not rooted in systematic inconsistencies.

For label ranking, however, both intra- and inter-annotator agreement drop substantially. The results show that even the same annotators would often change their ranking decisions when re-assessing the same sample, and imply that the ranking task is less stable and more subjective than the classification task. In summary, while the annotators align in their choice of labels in the classification task, there is considerably more uncertainty regarding how those labels should be ordered in the ranking task.

Ranking Confidence Besides ranking the labels, the annotators provide a self-assessed confidence score from 1 to 5 reflecting how certain they are in their ranking decisions. Evidently, annotators felt more confident in the second annotation round, likely because they were already familiar with the task and the articles’ style. As shown in Table 4.5, confidence levels are generally high; across all annotations, only four articles’ ranking component received a confidence score of 3, with all other ratings being either 4 or 5. Due to the limited number of annotated samples, we do not exclude the articles with lower confidence scores from the quantitative evaluation, but take them into consideration in the manual analysis. Nonetheless, two of the four articles with lower ranking confidence scores achieve full classification and near-perfect ranking agreement; one has an agreement score of 0.0 (entirely different), and the fourth one was re-annotated with a re-annotation overlap of 1.0 RBO. We can therefore conclude that there is no clear correlation between annotators’ self-reported confidence and their level of agreement.

Annotator	A0	A1	A2	A3	A4
Round 1	4.27	4.77	3.94	4.11	4.0
Round 2	4.5	5.0	4.75	5.0	4.0

Table 4.5: Averaged annotators’ confidence scores (self-reported) for the ranking task.

Comparison Between Models and Annotators We compare the performance of the zero-shot LLMs on the clean, original articles with human annotators to evaluate how closely automated predictions align with human judgment. We report performance across both tasks using Jaccard similarity to measure classification overlap and RBO/RBOx to assess ranking agreement.

Table 4.7 summarises the average agreement between the evaluated models and human annotators in the multi-label classification task. Overall, GPT-OSS 20B achieves the strongest performance across most metrics, with the highest Jaccard index (0.71) and F1-score (0.8) and second-highest recall of 0.88, which indicates that it captures most of the labels selected by the annotators. Mistral-Large 2 performs similarly and achieves the highest precision (0.83) and a high recall, which also show very close alignment with human evaluation. Llama3.3 70B has the highest recall (0.89) but a comparatively lower precision (0.67), meaning that it tends to over-select labels. Llama3.2 3B has the lowest agreement across all metrics and only aligns with the annotators to a limited extent.

In terms of ranking similarity (Table 4.6), Llama3.3 70B achieves the highest agreement with human annotators, reaching the best RBO (0.25) and RBOx (0.24) scores, followed closely by GPT-OSS 20B with 0.25 and 0.23. Mistral-Large 2 shows moderate performance, while Llama3.2 3B is the weakest overall. However, even with the larger models, the RBO and RBOx values remain very low, and suggest that none of the models orders the labels in a way that follows human annotators’ ranking patterns. To summarise, while some models are able to effectively identify relevant labels, compared to human judgment, they struggle to match human preferences in ranking them.

Metric	Mistral-Large 2	Llama3.3 70B	GPT-OSS 20B	Llama3.2 3B
RBO	0.220	0.254	0.245	0.173
RBOx	0.210	0.235	0.233	0.155

Table 4.6: Comparison of label ranking performance between human annotators and models (aggregated).

Metric	Mistral-Large 2	Llama3.3 70B	GPT-OSS 20B	Llama3.2 3B
Jaccard Index	0.684	0.611	0.706	0.472
Precision	0.828	0.665	0.774	0.532
Recall	0.781	0.891	0.879	0.815
F1-score	0.786	0.740	0.804	0.618

Table 4.7: Comparison of multi-label classification performance between human annotators and models (aggregated).

Finally, the manual error analysis reveals that human annotators and models have opposite ways of handling uncertainty: models would typically revert to over-selecting labels, while humans limit their predictions to very few labels when they are less certain in their assessment. Moreover, Llama3.2 3B frequently generates labels that are not included in the benchmark label set and produces inconsistently formatted outputs,

including symbols, numbers, and reasoning chains. These issues not only appear in the predictions from the clean articles, but also persist across all transformations. Surprisingly, the model is fairly consistent in selecting and ranking the same or semantically closely related labels that are not part of the pre-defined label set under transformations, which explains the inflated similarity scores despite its limited reliability.

5 Discussion

The following chapter addresses the key findings from each stage of the study and connects them to research on benchmarks and to developments in the climate domain. We first interpret the evaluation of the original benchmark and then discuss the redesigned tasks and their strengths and limitations.

Benchmark Exploration The benchmark analysis reveals the challenges of the single-label classification setup and suggests that evaluation based solely on agreement with the original labels may be a misleading marker for true model performance. While supervised approaches generalise better than zero-shot LLMs, their performance is still not optimal. The manual error analysis indicates that most misclassifications are not due to random error but rather to semantic and lexical overlap between categories, particularly in topics that are closely connected (Biology – Biotechnology) or have a hierarchical relationship (Zoology – Animals – Endangered Animals). This is further supported by the high inter-model agreement observed in the LLM evaluations on a subset, which shows a shared class interpretation among models that often differs from the dataset labels. Moreover, disagreement cases involve shifts from more specific categories to broader ones, such as Earthquakes being predicted as Geology or Weather as Climate.

These findings point to limitations in the benchmark design but can also be explained by the way the climate benchmark was constructed. As shown in Figure 5.1, category definitions in the Daily News are not static but change over time, with topics such as Ozone Holes, Genetically Modified, and Global Warming emerging or shifting according to current discourse trends. As a result, some labels become closely related or partially redundant, and models (and humans) can no longer systematically distinguish between them. As the dataset encompasses articles from a period of over 20 years, during which the political and social focus on climate change has evolved significantly, the definition and scope of many categories have also likely been transformed, reaching a state where there is no longer one true interpretation of the original classes.

Taken together, these factors contribute to a noisy, inconsistent benchmark with multiple overlapping classes that no longer reflect articles’ complexity and model performance in a meaningful way. Therefore, we suggest the task be reformulated as multiclass multi-label classification and label ranking with the aim to provide a more faithful representation of the underlying data and discourse shifts within the climate domain.

Multi-Label Classification and Ranking The results from the multi-label tasks, in both label classification and ranking, show great semantic stability across all variant pairs. The transformations are well aligned internally and trigger similar model behaviour despite conceptual differences. The code-mixed and the paraphrased variants test models’ dependency on keywords and superficial context cues, while the masked variants evaluate models’ susceptibility to information loss and diminished context availability. Unsurprisingly, as shown by the systematically lower masking scores across both tasks, models are more sensitive to content loss than to lexical changes. The results also hint that ranking is the more difficult task of the two under

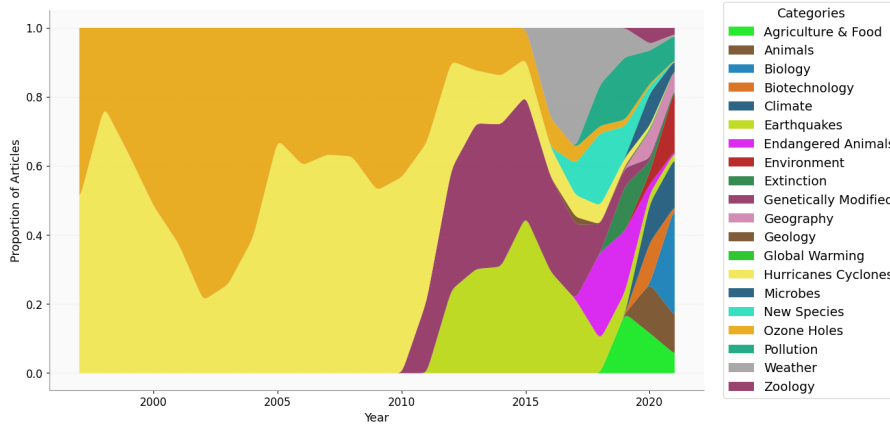


Figure 5.1: Evolution of Labels Over Time.

increased information reduction, and imply that accurately ordering labels requires broader access to the text’s general context than simply identifying them.

Contrary to the initial expectations of this study, the transformations produce only minimal variance in model outputs. The high degree of consistency suggests that the evaluated models are capable of ‘reading’ the articles even when the wording and structure of the text have been modified. From a robustness perspective, this is a positive finding, as it shows that the models’ predictions are not overly dependent on superficial lexical cues. However, the strong consistency across variants also raises questions about the effectiveness of the transformations in detecting true model performance. As the paraphrased and code-mixed transformations are designed to preserve the semantic meaning of the original text, they may not be sufficiently challenging for strong models like the ones we evaluate, and only partially introduce meaningful variation in performance. Therefore, while the transformations are semantically robust, they may be partially limited in capturing model vulnerability to perturbations. The exception to this is the masked variant, and particularly the articles transformed at the higher masking scales, as this modification appears to consistently cause performance degradation across both tasks. We can thus conclude that information loss is a more challenging perturbation than lexical variation and could be a more effective direction of future robustness testing.

That being said, the lexical transformations implemented in this study only cover relatively mild perturbation scenarios. For one, code-mixing is applied through mechanical word replacement with German translations. Considering that German is a very high-resource language that most modern LLMs are trained and optimised for, the multilingual variation is likely too accessible and straightforward to strongly impact model performance. Furthermore, the mechanical replacement strategy may reduce the natural complexity of real-world code-mixing, which often involves more irregular switching patterns (not just based on keywords as in our study) and contextual ambiguity. While we purposely apply a controlled transformation approach to ensure reproducibility and equal noise distribution across samples, it may represent code-mixing less naturally and not put enough stress on model robustness.

For the other, the paraphrased variants are generated by an LLM, which we prompt to rewrite the articles into youth slang. We intend to simulate a real-world setting, where scientific information appears on social media, but this adaptation may simplify the original text rather than add any linguistic complexity. Compared to the scientific style of the untransformed articles, the paraphrased variants reproduce the core contents but have a less complex structure (they are generally shorter), which could explain

their limited impact on model behaviour. Another factor may be the paraphrasing strategy, which uses a single LLM to transform the articles; the LLM (GPT5 Nano) may introduce certain stylistic biases or artifacts that our manual validation step does not detect but that are recognised by the evaluated models.

Although the variability across transformations remains low, the benchmark still captures differences in model ability. The performance variance between the stronger models and the weaker one is consistent throughout both tasks. In particular, Llama3.2 3B consistently underperforms compared to the larger models and has a predictable behaviour throughout all settings. At the same time, the marginal performance differences between the stronger models suggest that the benchmark may have limited discriminatory power at the upper end of performance evaluation. While it can reliably distinguish between weaker and stronger systems, it may require more challenging perturbations to better differentiate among the top-performing models. It is, however, also possible that the evaluated LLMs genuinely are robust to lexical variation and do, in fact, rely on deeper semantic representations rather than surface-level patterns, since they encounter similar challenges during training.

Although stronger models perform similarly across transformations, they do not produce the same outputs. The inter-model agreement analysis reveals that models do not fully converge on identical predictions – even models from the same architectural family – yet they do not select labels at random either. Instead, the moderate agreement scores suggest that the models share a common signal when analysing the articles and simultaneously adhere to model-specific classification and ranking patterns. This finding is particularly relevant for the ranking task: Although models often predict similar sets of labels, they differ substantially in how they order those labels. This raises important questions about the reliability of the ranking-based output assessment, as consistency across variants does not necessarily translate into agreement across models.

One more factor to consider is how ranking overlap is measured: The study uses rank-biased overlap (RBO), which is designed to capture differences between indefinite ranked lists, assigning greater weight to higher-ranked items. It has the property of assuming that observed ranking patterns persist in the indefinite space, i.e. that overlap behaviour in the ranked portion of the list continues similarly beyond the explicitly visible items. This assumption is useful for comparing incomplete or uneven ranked lists, but it may not fully reflect the actual disagreement distribution of the data, especially when models disagree on the order of lower-ranked labels beyond the top positions.

All factors considered, ranking outputs should be interpreted with greater caution than classification predictions, and the high scores we observe from most models in the ranking task under perturbations likely do not reflect true ranking performance. Without manual evaluation or comparison against human annotators, it remains difficult to determine whether the rankings indicate a ground truth or just model-specific ordering patterns.

It is for this reason that we adopt a manual evaluation approach, where we analyse a randomly selected subset of ranked lists across all models. The main finding here is that Llama3.2 3B appears less robust to (perturbed) inputs and more prone to generating outputs that are not compliant with the provided instructions and the defined label set. This suggests that smaller models are more limited in their ability to follow guidelines under constraints and noisy conditions. The recurring hallucinations, however, can artificially inflate the overall similarity scores, as the model repeatedly generates the same incorrect and out-of-set labels across all transformations, which makes it appear robust and consistent.

Additionally, both Llama models hack the evaluation algorithm by generating a higher number of labels in cases of uncertainty, sometimes up to 14 labels (out of 20 possible). In the examples we examine, the models select multiple labels that are only loosely related to keywords in the article and do not reflect its actual topics. Additionally, certain labels appear to be strongly connected in the models’ internal representations (e.g. Global Warming and Pollution) and are often predicted together even when only one of them is relevant. This behaviour naturally increases the probability of higher agreement between variants and automatically leads to higher similarity scores that may not reflect actual agreement. The longer sets of predicted labels have the added advantage of diluting the evaluation metrics’ effect: for the Jaccard similarity, even one mismatch in a shorter set could hurt the total score, whereas the total score of a longer set would barely be affected by that same mismatch. The RBO is not strict about set lengths either, and, as described above, if there is good alignment at the beginning of the ranking order, it assumes continuous similarity and does not penalise differences as much. Performance is, therefore, only partially covered by the metrics and, unfortunately, very susceptible to models’ gimmicks.

In contrast, human annotators adopt a more conservative strategy when they are less confident about the correct interpretation – they select only 1 or 2 labels. Unsurprisingly, this approach not only increases intra- and inter-annotator agreement, as fewer selected labels reduce the chances for disagreement, but it also contributes to lower agreement when comparing human-produced annotations to model predictions. As a result, the Llama models consistently achieve lower agreement with all individual annotators than the other models we evaluate.

Based on the inter-annotator agreement results, we can conclude that the multi-label classification task provides a sufficiently reliable foundation for benchmark evaluation, as annotators demonstrate strong consistency and a high level of label overlap. The label classification component can, therefore, be considered a valid measure of model performance. However, the much lower agreement observed in the ranking task suggests that it lacks stability and is more dependent on individual interpretation than on the predefined guidelines. Given this disagreement and the limitations in ranking evaluation, the task may need revision, particularly through merging semantically close labels that currently contribute to the disagreement among human annotators and models. The consolidation of overlapping labels into more distinct categories with strict boundaries could reduce ambiguity and improve consistency in both tasks, especially in the ranking task, where annotators often disagree on the relative ordering of closely related labels. However, such a revision could also decrease the level of granularity and the benchmark’s ability to detect context nuances, so one should carefully analyse the trade-offs before committing to such a major change.

Despite its limitations, the proposed benchmark offers great adaptability. Since news articles can be continuously collected through automated web scraping procedures, the dataset can be updated regularly to reflect emerging topics, research, and climate discourse trends. The transformation framework is also flexible and can be adapted as models continue to improve. If/when current perturbations become less challenging and obsolete, new strategies can be implemented to reflect model complexity and weaknesses. The benchmark has, therefore, the potential to be transformed into a dynamic benchmark that evolves along the requirements of the climate domain and the capabilities of modern LLMs by adapting its source material, tasks, and robustness testing methodology, to ensure a reliable long-term evaluation of model stability and reasoning under linguistic variation and fast-paced climate discourse shifts.

6 Conclusion

This study aimed to improve the Science Daily Climate Change benchmark for single-label topic classification by reviewing its original setup, identifying its limitations, and shifting the evaluation framework from single-label correctness to robustness and consistency under perturbations in multi-label classification and ranking tasks. The original benchmark was revised to better reflect the underlying ambiguity of the data and to reduce the need for reliable annotations of the entire dataset. To achieve this, three variants of the original climate articles were created: a paraphrased and a code-mixed variant that preserve the semantic meaning of the clean text but challenge models’ dependency on keyword patterns, and a masked variant with three scales of masking that tests models’ ability to derive context despite missing information. Four large language models of various sizes and architectures from three different model families were tested in a zero-shot setting across all transformations. The evaluation considered relative model performance, consistency, and robustness under perturbations, inter-model agreement, and alignment with human annotators on a subset of the data.

The results can be interpreted with regard to the three research questions as follows. Concerning RQ1, the redesigned multi-label classification task appears well-defined and provides a reasonable evaluation of model ability despite the large number of semantically overlapping labels and the inherent ambiguity of the articles. Annotators demonstrate high inter- and intra-annotator agreement, and based on the inter-model analysis, models mostly align in their predictions. The analysis further reveals that the task successfully discriminates between stronger and weaker models, but is limited in detecting differences in model behaviour among top-performing models, as the stronger LLMs achieve similarly high results across most evaluation metrics. However, the label ranking task shows a much lower inter-model and inter- and intra-annotator agreement, which suggests that the task may be more sensitive to the labelling noise and the data ambiguity.

Regarding RQ2, the results show that models’ predictions and relative performance remain strong and highly consistent across all transformations in both tasks, and only the masked variant with the most aggressive masking scale leads to performance degradation. Stronger models also exhibit greater stability across the perturbed inputs than the weaker model. The evaluated LLMs are therefore generally robust to input variation, and robustness increases with model size. However, the lexical transformations do not yield substantial performance differences among the stronger models, which indicates that the perturbations may not be sufficiently challenging to expose variance in LLMs’ ability to reason and generalise.

Finally, in relation to RQ3, the comparison between the annotated subset and models’ responses reveals that while models’ relative performance in the multi-label classification task is accurately represented by the benchmark, the strong robustness most models exhibit in the ranking task may be misleading. In the classification task, model predictions align closely with human annotations, whereas the ranking task yields substantially lower agreement between annotators and models. The subsequent analysis shows that models fail to converge towards a single ranking order, and some evaluation metrics’ properties contribute to artificially inflated final scores when models over-generate labels.

While the revised benchmark appears effective for evaluating multi-label classification, further improvements are needed to create a more reliable evaluation framework for complex multi-label ranking tasks. Future robustness testing may consider expanding the transformation strategies to include code-mixing with lower-resource languages from families unrelated to the Germanic branch, as well as paraphrasing that contains negation, perspective shifts, more natural social media language, or other forms of semantic-preserving but structurally disruptive paraphrases to test model robustness under more nuanced perturbations. In its current state, the benchmark is better suited for smaller or weaker models, as it can reliably distinguish between clear ability gaps. In a revised version of this study, however, we would focus on finding the upper bound of model performance on classification and ranking tasks that better differentiate among top-performing models.

Moreover, given the challenges in ranking evaluation, the inter-model and the inter- and intra-annotator disagreement, the ranking task may require further refinement in future versions of the benchmark. This could involve revision of the evaluation methods to account for models’ shortcuts, simplifying ranking criteria, or replacing strict ordinal ranking with alternative evaluation approaches that also rely on relevance. One further adjustment would be to reduce the number of labels by merging overlapping labels to form more distinct categories, which could reduce ambiguity and improve consistency across tasks.

Overall, this study contributes a revised SciDCC+ benchmark that better accounts for topic heterogeneity and introduces robustness testing through lexical transformations on a word and sentence level. Despite its limitations, the benchmark may serve as a basis for dynamic robustness evaluation that can be adapted according to current and future developments in language models and the climate-domain discourse.

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Appendix

6.1 Prompts

This section lists the prompts used to generate the paraphrased text variant and the prompts for zero-shot classification and ranking.

6.1.1 Paraphrased Variant Generation

GPT-5 Nano prompt:

Paraphrase the article in a non-scientific, Gen Z style. Rewrite the entire article from beginning to end and do NOT skip or add sentences. Keep grammar and formatting consistent. Output only the rewritten article. If the article is too long to process in one go, output “Null”.

6.1.2 Zero-Shot Single-Label Classification

This subsection contains the base prompt used for the zero-shot experiments as part of the original SciDCC single-label classification task. We evaluate the LLMs through two separate experiments – one with just the base prompt and one including the label definitions (the label definitions are listed in Section 6.3).

Base prompt:

You are a classifier. You are given an article (A) and a list of labels (L):

A: {article-body}

L: {labels}

Choose ONE label from the list of labels which matches the article’s topic best. Answer ONLY with the chosen label in JSON format. Do not produce any other output.

6.1.3 Zero-Shot Multi-Label Classification and Ranking

Before running the zero-shot experiments on the entire dataset, we conducted a small experiment with 200 instances to assess whether or not providing the label definitions we crafted helps the models generalise better. We tested across three models and all transformations. The expected outcomes were that the label definitions would increase consistency under perturbations (especially in edge cases) and that the outputs, where definitions were present, would be more closely aligned with human judgment, since the human annotators were provided with the same label definitions as part of the annotation guidelines. However, consistency remained identical, and the comparison with the annotations revealed no particular impact either. On the contrary, the models with definitions selected a larger number of labels in cases where humans would normally select 2 or 3. The manual analysis showed that models with definitions consistently over-predict closely related classes, such as Endangered Animals, Animals, and Zoology, or Climate, Environment, and Weather; the annotators have classified those examples only as Endangered Animals and Climate. Models

without the definitions given behave similarly, but predict fewer labels in said edge cases. Considering these findings and the increased cost of passing in longer prompts, we opted against the inclusion of label definitions in the final multi-label zero-shot prompts.

Multi-Label Classification

This subsection lists the prompts used for the zero-shot multi-label classification task. The base prompt is applied in the experiments with unperturbed articles and across all transformations, except for the code-mixed variant that uses an adjusted prompt to account for repetitive text strings that the postprocessing procedure might have missed.

General prompt:

You are a classifier. You are given an article (A) and a list of labels (L):

A: {article-body}

L: {labels}

Read through the article and carefully analyse its contents. Choose all labels that match the article's topic. There is no limit to the number of labels you may choose, as long as they are relevant to the article's subject(s). You may also assign only one label if that is the only topic of the article. Output ONLY the chosen labels in a JSON list. Do not produce any other output.

Code-mixed variant prompt:

You are a classifier. You are given an article (A) and a list of labels (L):

A: {article-body}

L: {labels}

Read through the article and carefully analyse its contents. Ignore long sequences of repetitive tokens, if any exist. Choose all labels that match the article's topic. There is no limit to the number of labels you may choose, as long as they are relevant to the article's subject(s). You may also assign only one label if that is the only topic of the article. Output ONLY the chosen labels in a JSON list. Do not produce any other output.

Multi-Label Ranking

This subsection lists the prompts used for the zero-shot multi-label ranking task. Similarly to the classification task, we apply a base prompt throughout all transformations, except for the code-mixed variant that uses an adjusted prompt.

General prompt:

You are a classifier. You are given an article (A) and a list of labels (L):

A: {article-body}

L: {labels}

Compare ALL labels from the list in PAIRS with each other (i.e., every possible combination) and evaluate, for each pair, which label describes the topic of article A better. From these comparisons, derive a full ranking. Output the labels in a JSON list in the order of their ranks, starting with the most relevant. Do not produce any other output.

Code-mixed variant prompt:

You are a classifier. You are given an article (A) and a list of labels (L):

A: {article-body}

L: {labels}

[Disclaimer: Ignore long sequences of repetitive tokens in the article, if any exist.]
Compare ALL labels from the list in PAIRS with each other (i.e., every possible combination) and evaluate, for each pair, which label describes the topic of article A better. From these comparisons, derive a full ranking. Output the labels in a JSON list in the order of their ranks, starting with the most relevant. Do not produce any other output.

6.2 Transformations

This section presents an example article in its original form and in all of its transformed versions.

Original Article

The observations help fill a gap in data on the seismic activity of the SAF in northern California, particularly south of San Francisco. Geologists Thomas Fumal and Tim Dawson conducted paleoseismic studies at Mill Canyon, near Watsonville, California. They documented evidence for four earthquakes, the most recent being the 1906 M 7.8 San Francisco event. They conclude that each of the three earthquakes prior to the 1906 quake was a large magnitude event that likely ruptured most, or all, of the Santa Cruz Mountains segment, producing similar physical deformation as the 1906 quake. In addition to filling in a data gap about the SAF in this region, this research adds to the understanding of how the SAF behaves, in particular, whether individual segments of the fault system can produce destructive earthquakes and how often. This study joins to a growing body of work that suggests the SAF produces a wider array of magnitudes than previously appreciated in the current seismic hazard models.

Code-Mixed Article

Die Beobachtungen helfen fill a gap in data on the seismische Aktivität of the SAF in Nordkalifornien, particularly south of San Francisco. Geologen Thomas Fumal und Tim Dawson durchgeführte paläoseismische Studien at Mill Canyon, near Watsonville, California. Sie dokumentierten Beweise for four earthquakes, the most recent being the 1906 M 7.8 San Francisco event. They conclude that each of the three Erdbeben vor to the 1906 quake was a Großereignis that likely ruptured most, or all, of the Santa Cruz-Gebirge segment, producing similar physical deformation as the 1906 quake. In addition to filling in a data gap about the SAF in this region, this research adds to the understanding of how the SAF behaves, in particular whether individual segments of the fault system can produce zerstörerische Erdbeben and how often. This study joins to a growing body of work that suggests the SAF produces a wider array of magnitudes than previously appreciated in the aktuell seismisch hazard models.

Paraphrased Article

Basically, these findings filled in missing tea about earthquake activity on the SAF in Northern California, especially south of San Francisco. Geologists Thomas Fumal and Tim Dawson studied old quake evidence at Mill Canyon near Watsonville, California. They found proof of four earthquakes, with the latest being the massive 1906 San Francisco quake at magnitude 7.8. They think the three quakes before 1906 were also giant events that probably ripped through most or all of the Santa Cruz Mountains section, causing similar destruction and ground changes as the 1906 quake. Besides filling a data gap about the SAF in this area, the research also helps explain how the fault behaves, like whether one section of the fault can randomly go full menace mode with a destructive quake and how often that happens. This study adds to growing research showing the SAF can create a wider range of earthquake magnitudes than older seismic hazard models originally thought.

Article at 0.25 Masking

The observations help fill a gap in data on the <MASK> of the SAF in northern California, particularly south of San Francisco. Geologists Thomas Fumal and Tim Dawson conducted paleoseismic studies at Mill Canyon, near Watsonville, California. They documented evidence for four <MASK>, the most recent being the 1906 M 7.8 San Francisco event. They conclude that each of the three <MASK> to the <MASK> was a large magnitude event that likely ruptured most, or all, of the Santa Cruz Mountains segment, producing similar physical deformation as the <MASK>. In addition to filling in a data gap about the SAF in this region, this research adds to the understanding of how the SAF behaves, in particular, whether individual segments of the fault system can produce <MASK> and how often. This study joins to a growing body of work that suggests the SAF produces a wider array of magnitudes than previously appreciated in the current <MASK> models.

Article at 0.5 Masking

The observations help fill a gap in data on the <MASK> of the <MASK> in <MASK>, particularly south of San Francisco. Geologists Thomas Fumal and Tim Dawson <MASK> studies at Mill Canyon, near Watsonville, California. They <MASK> for four <MASK>, the most recent being the 1906 M 7.8 San <MASK>. They conclude that each of the three <MASK> to the <MASK> was a large magnitude event that likely ruptured most, or all, of the Santa Cruz Mountains segment, producing similar physical deformation as the <MASK>. In addition to filling in a data gap about the <MASK> in this region, this research adds to the understanding of how the <MASK> behaves, in particular, whether individual segments of the fault system can produce <MASK> and how often. This study joins to a growing body of work that suggests the <MASK> a wider array of magnitudes than previously appreciated in the <MASK> hazard models.

Article at 0.75 Masking

The observations help fill a gap in data on the <MASK> of the <MASK> in <MASK>, particularly south of San Francisco. Geologists Thomas Fumal and Tim Dawson <MASK> studies at Mill Canyon, near Watsonville, California. They <MASK> for four <MASK>, the most recent being the 1906 M 7.8 San <MASK>. They conclude that each of the three <MASK> to the <MASK> was a large magnitude event that

likely ruptured most, or all, of the Santa <MASK> segment, producing similar physical deformation as the <MASK>. In addition to filling in a data gap about the <MASK> in this region, this research adds to the understanding of how the <MASK>, in particular, whether individual segments of the fault system can produce <MASK> and how often. This study joins to a growing body of work that suggests the <MASK> a wider array of magnitudes than previously appreciated in the <MASK> <MASK>.

6.3 Annotation Guidelines and Label Definitions

This section presents the annotation guidelines and the label definitions we composed as part of the guidelines to improve classification and ranking consistency. The label definitions are also included in selected zero-shot experiments.

Annotation Guidelines

You are given a short scientific article and a set of 20 labels. You have two tasks. Please complete them one by one:

Task 1.

Read through the article and analyse its contents. From the set of labels, choose all labels that match the article's topic based on the labels' definitions below. There is no limit to the number of labels you may choose, as long as they are relevant to the article's subject. You may also assign one label if that is the only topic of the article. Once you have decided on the label(s), proceed to task 2.

Task 2.

Rank the labels you have chosen based on their explicit relevance to the article's topic(s), starting with the most relevant/important/explicit/descriptive label. Be sure to pay attention to how different subjects within the same article are connected and what the main article's insights are. When in doubt, you may use visual markers, such as the length of text dedicated to one topic, to form a decision. If you encounter tricky cases, please write down your thoughts in the comments column.

Task 2.1.

Lastly, please provide a score from 1 to 5 on how confident you are in your ranking assessment:

- 1: not confident at all
- 2: slightly confident
- 3: somewhat confident
- 4: fairly confident
- 5: completely confident

Write down your answers, following the examples in the next tab. Please preserve the formatting of the labels (incl. special symbols or lack thereof, capital letters in the beginning, etc.). If you encounter any tricky cases or have feedback, write down your thoughts in the comments column.

Label Definitions

- Agriculture & Food - the science, art, or practice of cultivating the soil, producing crops, and raising livestock; in varying degrees, the preparation and marketing of the resulting products; nutriment in solid form that people and animals eat, or plants absorb, to keep them alive;
- Animals - any of a kingdom (Animalia) of living things, including many-celled and single-celled organisms, that typically differ from plants, require more complex food materials, and are organised to a greater degree of complexity;
- Biology - the scientific study of the natural processes of living organisms; the plant and animal life of a region or environment; the life processes, especially of an organism or group;
- Biotechnology - the manipulation of living organisms or their components to produce useful, usually commercial products; any of various applications of biological science used in such manipulation;
- Climate - the general weather conditions usually found in a particular place;
- Earthquakes - sudden, violent movements of the earth's surface, sometimes causing great damage;
- Endangered Animals - animals threatened with extinction;
- Environment - the complex of physical, chemical, and biotic elements that exist in nature;
- Extinction - the condition or fact of being extinct; the event in which one or more species of living things have or will become extinct;
- Genetically Modified - (ex. animal, plant) has had some of its genes changed scientifically;
- Geography - science that deals with the description, distribution, and interaction of the diverse physical, biological, and cultural features of the earth's surface;
- Geology - the study of the rocks and physical processes of the Earth in order to understand its origin and history;
- Global Warming - the gradual increase in the Earth's temperature, caused by gases, esp. carbon dioxide, surrounding the Earth.
- Hurricanes Cyclones - violent storms with strong circular winds;
- Microbes - organisms (such as bacteria) of microscopic or ultramicroscopic size;
- New Species - a class of individuals having common attributes and designated by a common name that has recently come into existence;
- Ozone Holes - areas of the ozone layer that are depleted of ozone;
- Pollution - damage caused to water, air, etc. by harmful substances or waste; the act of polluting, or the substances that pollute;

- Weather - the conditions in the air above the earth, such as wind, rain, or temperature, especially at a particular time over a particular area;
- Zoology - the scientific study of the behaviour, structure, physiology, classification, and distribution of animals;

6.4 Multi-Label Classification Consistency across Variants

Figure 6.1 shows the pairwise Jaccard similarity across all transformation pairs for each evaluated model. Overall, the results indicate that responses remain highly consistent across perturbations, particularly for the larger models. Mistral-Large and Llama3.3 70B exhibit the strongest stability, with most similarity scores remaining close to or above 0.85 across variants. GPT-OSS 20B shows moderately high consistency, while Llama3.2 3B has noticeably lower similarity values, which suggests that it is more sensitive to the changes in input. Across all models, closely related masked variants retain relatively high similarity and diverge more strongly as the masking scale increases.

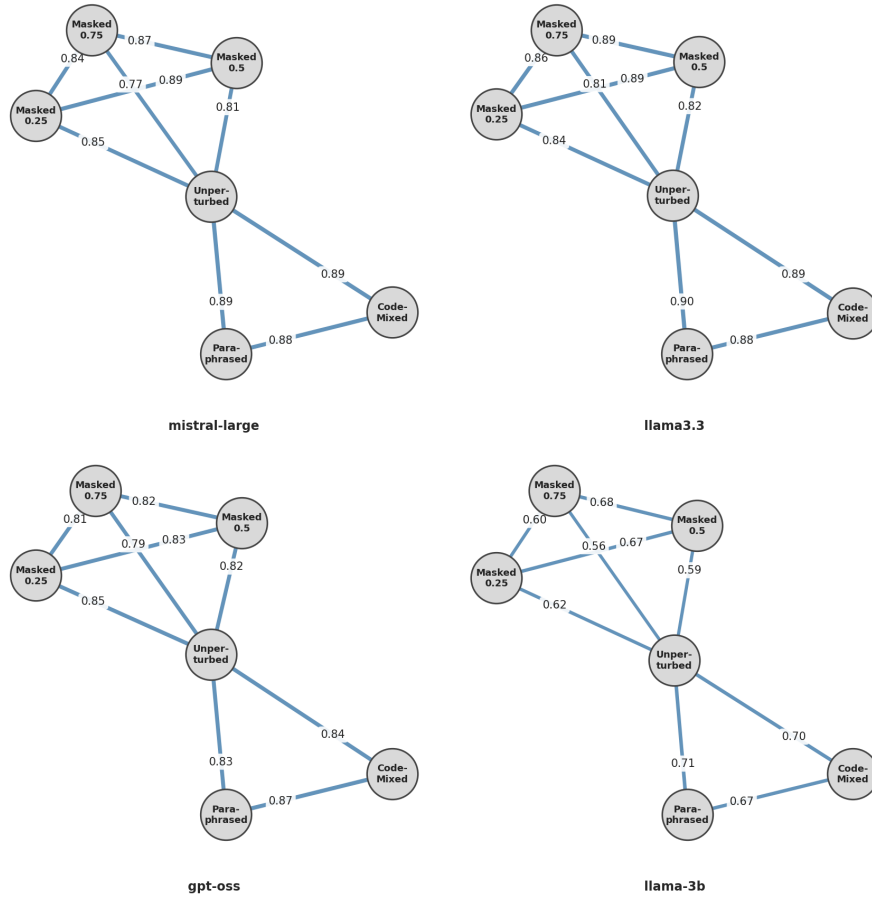


Figure 6.1: Consistency across variants.

6.5 Model-Annotator Agreement Analysis

In this section, we present a more fine-grained comparison of model predictions and annotator judgment for the multi-label classification (Table 6.1) and ranking (Table

Annotator	Metric	Mistral-Large 2	Llama3.3 70B	GPT-OSS 20B	Llama3.2 3B
A0	Jaccard Index	0.597	0.584	0.649	0.425
	F1-score	0.709	0.724	0.756	0.581
	Precision	0.782	0.648	0.772	0.491
	Recall	0.693	0.893	0.804	0.782
A1	Jaccard Index	0.716	0.649	0.717	0.517
	F1-score	0.822	0.778	0.811	0.661
	Precision	0.881	0.701	0.775	0.585
	Recall	0.791	0.913	0.881	0.843
A2	Jaccard Index	0.748	0.686	0.771	0.519
	F1-score	0.833	0.799	0.851	0.653
	Precision	0.857	0.718	0.825	0.576
	Recall	0.835	0.950	0.898	0.820
A3	Jaccard Index	0.652	0.554	0.685	0.406
	F1-score	0.755	0.694	0.793	0.558
	Precision	0.801	0.628	0.767	0.454
	Recall	0.739	0.830	0.862	0.794
A4	Jaccard Index	0.709	0.583	0.706	0.494
	F1-score	0.812	0.707	0.810	0.636
	Precision	0.820	0.631	0.730	0.556
	Recall	0.847	0.867	0.950	0.836

Table 6.1: Comparison of multi-label classification performance between human annotators and models.

Annotator	Metric	Mistral-Large 2	Llama3.3 70B	GPT-OSS 20B	Llama3.2 3B
A0	RBO	0.176	0.228	0.201	0.180
	RBOx	0.164	0.208	0.190	0.159
A1	RBO	0.277	0.261	0.297	0.167
	RBOx	0.268	0.244	0.283	0.151
A2	RBO	0.225	0.278	0.259	0.172
	RBOx	0.217	0.259	0.249	0.157
A3	RBO	0.204	0.248	0.226	0.164
	RBOx	0.195	0.228	0.214	0.144
A4	RBO	0.215	0.257	0.243	0.183
	RBOx	0.205	0.238	0.227	0.165

Table 6.2: Comparison of label ranking performance between human annotators and models.

6.2) tasks. We compare each model individually against each annotator to examine whether certain models align more closely with particular annotators.

In the multi-label classification task, Annotator 2 (A2) consistently achieves the strongest agreement with all evaluated models, with the highest Jaccard and F1-scores across all four models. Furthermore, annotators show the highest agreement with GPT-OSS 20B and Mistral-Large 2, which suggests that these models most closely reflect human judgment. In the ranking task, the agreement patterns are more mixed, although Annotator 1 (A1) achieves the highest ranking similarity with predictions from half of the models, including Mistral-Large 2 and GPT-OSS 20B. Most annotators (3 out of 5) appear to have their highest agreement with Llama3.3 70B. Across both tasks, Llama3.2 3B consistently records the lowest agreement scores, showing weaker alignment with human annotations overall.